INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN SOUTH AFRICA

Document Prepared by

C-Quest Capital CR Stoves Pte. Ltd.

Project Title	Installation of high efficiency wood burning cookstoves in South Africa
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1 PROJECT DETAILS

1.1 Summary Description of the Project

A summary description of the technologies/ measures to be implemented by the project.

The project status is under development. The project involves distribution of fuel-efficient improved cookstoves (ICS) in South Africa. The ICS disseminated through this project will replace the baseline cookstoves i.e. open fire or three stone fire or tripods using firewood or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney

Through this project, the distribution and installation of approximately 500,000 ICS will be undertaken for about 250,000 households in Republic of South Africa. It is intended that under this project a single pot, TLC-CQC Rocket Stove will be distributed, or any other fixed or portable ICS model may be added based on the requirement of end users as suitable. Each household will receive up to 2 ICS per household[1].

The end user will be informed in advance that the use of ICS generates carbon finance which in turn is used for subsidising the price of ICS and for recovering project implementation costs. At the time of validation, no ICS were distributed under the grouped project activity.

The Location of the Project.

The project will take place in Republic of South Africa. The details of the project location are provided in Section 1.12.

An explanation of how the project is expected to generate GHG emission reductions or removals.

The ICS will substitute the currently common cooking on open fire or three stone fired stoves or a conventional device with no improved combustion air supply or flue gas ventilation that is without a grate or a chimney. The ICS burns wood more efficiently thereby improving thermal transfer to pots, hence saving fuel and lowering greenhouse gas emissions. This not only will reduce the rapidly progressing deforestation in Republic of South Africa but will also reduce health hazards from indoor air pollution and women and children will have to spend less time collecting firewood.

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[1] CQC has implemented ICS projects with single ICS for HHs in its previous projects (PoA 9007, PoA 9558, PoA 8060), where it was observed that a single stove is insufficient for the household requirements and the end users continued to use the traditional stoves along with the project ICS. This led to a more practical and sustainable two-stove distribution initiative per household by CQC. This approach will ensure that the end users have complete access to clean cooking solutions and that the devices are sufficient for their cooking requirements, ensuring that the household doesn't go back to the baseline practice of three-stone fired/tripod stoves.

Even recent projects implemented by CQC (VCS 2521, 2522, 2523) have adopted the same approach followed by the current project activity too.

A brief description of the scenario existing prior to the implementation of the project.

The current scenario in the baseline is cooking on open fire or three stone fired stoves or a conventional device with no improved combustion air supply or flue gas ventilation that is without a grate or a chimney. The ICS burns wood more efficiently thereby improving thermal transfer to pots, hence saving fuel and lowering greenhouse gas emissions.

An estimate of annual average and total GHG emission reductions and removals.

The average annual GHG emission reductions from the project are expected to be 1,004,676 t CO_2e. Total GHG emission reduction from the project is expected to be 10,046,761 tCO_2e over the entire crediting period of 10 years.

Audit Type	Period	Program	VVB Name	Number of years
Validation	01-February-2024[2]	VCS	Carbon Check (India) Private Limited	10 (Fixed crediting period)

1.2 Sectoral Scope and Project Type

The project is categorised under type/category as below:

a) **Sectoral scope**: 03 - Energy demand
b) **Type:** II – Energy efficiency improvement projects

The project is a grouped project.

1.3 Project Eligibility

As per para. 2.1. of the VCS Standard, V.4.4, the project is eligible under the scope of the VCS Program because the VCS Scope includes project activities involving the following elements:

Table 1: VCS Programme scope

As per para 2.1 of VCS standard Version 4.4	
Items vide para 2.1.1 of VCS standard V4.4	**Applicability**
1 The six Kyoto Protocol greenhouse gases	The project activity reduces GHG emissions associated with the combustion of cooking

[2] Expected start date of project activity and same can be verified during first periodic verification by the Verifying VVB.

	fuels. The project activity aims at CO_2 reduction of non-renewable woody biomass.
2 Ozone-depleting substances.	The project does not involve any zone depleting substances as project cooks stoves uses woody biomass.
3 Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	The project activity is distribution of improved cookstoves, "VMR0006 "Methodology for Installation of High Efficiency Firewood Cookstoves", version 1.1." which is a VCS approved methodology is applicable for the project activity.
4 Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Not applicable
5 Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements	Not applicable
Items vide para 2.1.2 of VCS standard 4.4	**Applicability**
The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.	Not applicable as the project is voluntary and GHG emissions are not for the purpose of their subsequent reduction, removal, or destruction.
Items vide para 2.1.3	**Applicability**
The VCS Program also excludes the project activities under the circumstances indicated in Table 1, of VCS Standard 4.4.	The project activity efficiency improvements in thermal applications (e.g., cook stoves) is not listed in the category of excluded project activities table in section 2.1.3 of the VCS Standard ver 4.4

Thus, the project activity falls within the scope of the VCS Program and is eligible under the scope of the VCS Program.

1.4 Project Design

The project is a grouped project.

Eligibility Criteria

For the inclusion of new project activity instances, the project proponent will ensure that it meets the eligibility criteria according to section 3.6.16 and 3.6.17 of the VCS Standard 4.4, as stated in the below Table.

Table 2: Eligibility criteria as per VCS standard (Version 4.4)

S.No.	Criterion	Compliance Requirement	Method of Evaluation
1	**Methodology:** Meet the applicability conditions set out in the methodology applied to the project	The project activity instance shall use VCS approved methodology-VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1 and shall meet the applicability conditions set out in the methodology.	Details of how the new project activity instances meet the applicability conditions can be confirmed in Section 3.2 of this VCS-PD.
2	**Technology:** Use the technologies and measures specified in the project description	The project activity instance shall implement TLC-CQC Rocket Stoves having minimum efficiency of 25%; below which it shall not be eligible under this grouped project.	Thermal Efficiency test report/ manufacturer 's specification shall be used for establishing minimum efficiency criteria in the first year of stove installation.
3	**Baseline Scenario:** Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All new project activity instances will be installed within Republic of South Africa only and the baseline scenario is the continued use of non-renewable wood fuel by the target	Baseline scenario can be confirmed in Section 3.4 of this VCS-PD. Baseline scenario and geographical boundary will be confirmed from registration records cum consent deeds of individual ICS which will

		population to meet similar thermal energy needs as provided by the project cookstoves in absence of the project activity. This also confirms to the applied methodology.	contain geographical coordinates, address as well as confirmation by end user (stove owner) stating that they were using traditional three stone fire or pot support before the installation of project stove.
4	**Additionality:** Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	Each project activity instance shall demonstrate its compliance with- **1.Regulatory Surplus:** The project should not be mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework. **2. Positive List:** The inclusion of new project activity instances shall have to comply with positive list that is the ICS are installed/ distributed at zero cost to end user and the project activity instance has no other source of revenue but GHG credits.	1. There is no law, statute or government programme or policy in Republic of South Africa mandating the project activity nor is there any systematically enforced law, statute or other regulatory framework for such projects. 2. Self-Declaration/individual ICS registration form or any other documentary evidence mentioning that end users were not charged for the ICS.
5	Occur within one of the **designated geographic areas** specified in the project description	All new project activity instances will be installed within Republic of South Africa only.	Project Activity Instance Database containing Geographical Coordinates of each household to which cookstoves are distributed.

6	**Ownership:** Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance	A default Beneficiary Agreement for end users including the provision that emission reductions generated by the project activity are owned by the Project Proponent will be provided for project activity instance.	Copy of registration form cum consent deed from end user (Stove owner) to Project Proponents regarding emission reduction claims.
7	**Start Date**	The start date of project activity instance shall be same as or later than the grouped project start date	Project Activity Instance Database containing date of installation of each ICS.
8	Where a **capacity limit** applies to a project activity included in the project, no project activity instance shall exceed such limit	1. The aggregate energy savings by a single project activity shall not exceed the equivalent of 180 GWh_{th}/year as mentioned in the methodology AMS II G: Energy efficiency measures in thermal applications of non-renewable biomass. 2. Each project activity instance that exceeds one percent of the capacity limit (i.e., 1.8 GWh_{th}) shall be identified. 3. Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the	1. Project Activity Instance Spreadsheet containing estimation of annual energy saving from each project activity instance (i.e., each project ICS). 2. Project Activity Instance ER Spreadsheet containing estimation of annual energy saving from each project activity instance (i.e., each project ICS). 3. The expected annual energy saving from each project activity instance (i.e., each project ICS) comes out to be 0.009 GWh_{th}/ TLC-CQC Rocket Stove which is well below 1% (i.e., 1.8 GWh_{th}) of the capacity limit. Therefore, the project activity instances under this grouped project shall not be assigned to clusters.

		cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters.	
9.	**Double Counting**	The Improved Cookstove distributed under this project shall be uniquely identifiable based on the distribution records	Project Activity Instance Database containing unique serial number of each ICS. Each ICS distributed will have corresponding end user (stove owner) details (i.e., name, geographical coordinates, address, unique serial number). The combination of this unique serial number of each ICS along with geographical address of the stove owner confirms the unique identity of each project device. If there is another implementing partner added in the project in future, PP will ask for declaration from implementing partner that that they will not seek GHG emission claim in any other standard.
10	**Target Population**	End user for each project activity instance shall be households, with non- renewable biomass on inefficient wood stoves. Each ICS will be assigned a unique serial number with name of ICS user, address, GPS of household, stove model, distribution date, etc.	Target population will be the households relying on non-renewable woody biomass on traditional three stone or open fire. Same can be confirmed from Database/ICS Registration form if distribution has started or Self- declaration by project participant.
11	**Crediting period**	Be eligible for crediting from the start date of the instance through to the end of the project	The start date of the instance will be the date the stove began generating GHG emission reductions or removals. The end of the

		crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4) and new instances are eligible for crediting from the start of the next verification period.	crediting period of instance will be the end date of project crediting period as mentioned in section 1.9.

1.5 Project Proponent

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Contact person	Ken Newcombe
Title	Director
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Email	cqc-operations@cquestcapital.com

1.6 Other Entities Involved in the Project

At present, C-Quest Capital CR Stoves Pte. Ltd. is the sole entity involved in the project.

1.7 Ownership

The project ownership is with C-Quest Capital CR Stoves Pte. Ltd.

Further, Registration certificate cum consent deed signed by end user (project stove owner) to PP during the registration process includes the statutory clause “Acknowledgement and Carbon Rights Waiver – My household hereby transfers and waives all and any rights in connection with the improved cookstove or project and greenhouse gas emission reductions or other environmental or social attributes generated by the installation and use of the improved cookstove to C-Quest Capital CR Stoves Pte. Ltd. and we understand that C-Quest Capital CR Stoves Pte. Ltd. may, at any time, transfer those rights to an affiliate without further notice or consent".

1.8 Project Start Date

01-February-2024 (The estimated date of commissioning of first ICS under this grouped project activity)

1.9 Project Crediting Period

01-February-2024 to 31-January-2034, 10 years fixed crediting period

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are >300,000 tonnes of CO_2e per year and hence project falls under large project as per 3.10.1 of VCS standard version 4.4.

Project Scale	
Project	
Large project	X

Year	Estimated GHG emission reductions or removals (tCO_2e)
01-February-2024 to 31-January-2025	1,907,803
01-February-2025 to 31-January-2026	1,692,197
01-February-2026 to 31-January 2027	1,482,329
01-February-2027 to 31-January-2028	1,278,114
01-February-2028 to 31-January-2029	1,079,467
01-February-2029 to 31-January-2030	886,308
01-February-2030 to 31-January-2031	698,553
01-February-2031 to 31-January-2032	516,124
01-February-2032 to 31-January-2033	338,941

01-February-2033 to 31-January-2034	166,925
Total estimated ERs	**10,046,761**
Total number of crediting years	**10**
Average annual ERs	**1,004,676**

1.11 Description of the Project Activity

The project involves distribution of fuel-efficient improved cookstoves (ICS) to replace[3] three-stone fires or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system i.e., traditional stoves in households. The baseline scenario is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of grouped project.

Introduction of ICS as energy efficiency measure:

The ICS improves the efficiency of combustion and thermal transfer to the pot compared with a traditional pot support or three-stone fire because of improved design which leads to efficient combustion of wood. This substantially reduces wood fuel consumption compared with a three stone fire or traditional pot support.

The geographical boundary is in Republic of South Africa. Under this project, each household will receive a maximum of two units of TLC-CQC rocket stoves, or any other fixed or portable ICS model may be added based on the requirement of end users as suitable. In case, the stove design and specifications are different from the current stove, the technical details of project stove, including specifications and efficiency rates will be included in the monitoring report of the respective monitoring period. The project activity will generate greenhouse gas (GHG) emission reductions by avoiding CO_2 emissions as a function of the reduction in the amount of non- renewable woody biomass fuel consumption in the efficient project stoves as compared to the baseline stoves.

Technology/Measure

TLC-CQC Rocket Stove uses a total of 15 bricks that will be made by the households using locally available clay. The average size of the brick would be 22cm x 11cm x 6.5cm. The bricks

[3] For the baseline cookstove, validation VVB has raised a FAR, thereby during the first verification verifying VVB will check the usage of the baseline cookstove.

will be joined together using a mixture of 5 liters clay, 5 liters sand, 5 liters manure/cow dung and 5 liters of water. This ensures reduction in heat loss and better insulation. Metal components have been added to the design to optimize combustion and heat transfer.

Figure 1: TLC-CQC Rocket Stove

Stove components

The stove has a metal top that allows the pot to sit higher, improving the flow of air into the combustion chamber and out through the top of the stove.

An adjustable metal pot skirt ensures more effective transfer of heat from the fire into the pot, increasing efficiency and helping to block wind.

TLC-CQC rocket stoves has a metal stick support which is placed in front of and slightly into the opening of the stove and acts as a firewood feeding platform. This ensures adequate airflow while feeding the fuel into the combustion chamber resulting in complete combustion of wood.

According to independent stove efficiency tests performed by Aprovecho Research Centre on the TLC-CQC- rocket stove, the WBT results yielded an average thermal efficiency of 34.5%.

Table 3: Details of project Stove (TLC-CQC rocket stove)

Description	Technical Specifications
Stove Size	Depth: 35 cm
	Width: 35 cm
	Height: 28 cm
Combustion Chamber Size	Depth: 12 cm
	Width: 12 cm
	Height: 28cm
Stove high-power thermal efficiency	34.5%
Average Lifespan	10 Years

The life of TLC-CQC rocket stove is at least 10 years owing to it being a clay and brick stove which can be easily repaired using a mixture of local clay, cow dung, sand, and water on a regular basis. The metal parts used in the stoves are under extended warranty from PP/manufacturer throughout the projected life span and will be replaced in case of request received from the stove owner for replacement. Thereby confirms the life span of 10 years through the above measures.

Data collection of ICS end-user

Project proponents must gather the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- Stove model
- Date of distribution/installation
- ICS serial number
- Retailer/distributer information

The information collected will be stored in the electronic database which will serve as a project database for project monitoring and sampling purposes.

1.12 Project Location

The project boundary for the grouped project is geographical boundary of Republic of South Africa with coordinates 30°33' 34.1'' S latitude and 22°56.25' E longitude.[4]

[4] https://www.geodatos.net/en/coordinates/south-africa

Figure 2: Republic of South Africa map[5]

Figure 3: Map of project area[6]

[5] https://www.globalsecurity.org/military/world/rsa/maps.htm

The project boundary for each Project Instance will be defined within the geographic borders of the Republic of South Africa.

1.13 Conditions Prior to Project Initiation

The condition prior to project implementation is the continued of three stone fired cookstoves or a conventional device with no improved combustion air supply or flue gas ventilation (that is without a grate or a chimney) by the target population by burning non-renewable woody biomass to meet the thermal energy needs as provided by project cookstoves in absence of project activity. Also, more information is provided in section 3.4 below.

It is a common and traditional practice in the project area to use three stone fired stoves or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. Hence, the applicable efficiency fraction for three stone fired or conventional stoves which stands at 0.1 (default) as per the applied methodology VMR0006 1.1 is considered in the baseline estimations.

Baseline related information is provided to VVB during validation.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project is a voluntary initiative by the project proponent and has not been implemented to meet any local/national laws or regulatory compliances.

A review is made on South Africa environmental laws and regulations as below:

a) The National Forests Act (Act 84 of 1998)[7]

b) National Environmental Management Act, 1998 (Act 107 of 1998)[8]

c) National Environmental Management, Protected Areas Act (Act 57 of 2003)[9]

d) National Environmental Management, Biodiversity Act (Act 10 of 2004)[10]

e) Environmental Impact Assessment Regulations, 2014[11]

There is no negative concern made on improved cookstoves project from the above laws and regulations.

1.15 Participation under Other GHG Programs

[6] KML file provided to VVB as a part of validation.

[7] https://www.gov.za/sites/default/files/gcis_document/201409/a84-98.pdf

[8] https://www.gov.za/documents/national-environmental-management-act

[9] https://www.dffe.gov.za/sites/default/files/legislations/nema_amendment_act57.pdf

[10] https://www.dffe.gov.za/sites/default/files/legislations/nema_amendment_act10.pdf

[11] https://cer.org.za/wp-content/uploads/2010/03/EIA-Regulations-2014.pdf

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project is not included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related credit, including renewable energy certificates.

Supply Chain (Scope 3) Emissions

As per Section 3.23 of the VCS Standard, v4.4, the "producer(s) or retailer(s) of the impacted good or service are known but not involved in the project or do not have a website".

The Project Proponent will inform the manufacturers of ICS as well as the implementing partner at the time of signing agreement for the goods and services provided by them, that ICS will be distributed with in the project boundary (Republic of South Africa) under this project activity. Also, will intimates them about the project registration under VERRA for the generation of VCUs and the risk of double claiming.

Evidence of the public statement(s) to avoid double counting by PP on their website under public notice[12] for Scope 3 emission has been provided as Appendix 2.

As implementation begins PP will ensure notifying the project partners on the possible conflict situation with Scope 3 emissions double claiming. Respective evidence will be provided to the verifying VVB during first verification.

1.17 Sustainable Development Contributions

The project contributes to sustainable development in a number of ways:

a) Environmental Sustainability

[12] https://cquestcapital.com/latest/public-notices/

- The project will help significantly reduce greenhouse gas emissions over its lifetime.
- The project will help reduce the use of non-renewable woody biomass from forests, thus assist in conserving existing forest stock and the protection of natural forest eco-systems and wildlife habitats.

b) Social Sustainability

- The project stoves will considerably reduce the time needed to be spent in collecting wood fuel and cooking activities of the household thereby reducing the work burden on rural families and presenting alternative opportunities for economic development.
- The amount of indoor pollutants from burning of woody biomass in the household will be reduced. Less carbon dioxide, carbon monoxide and particulates will be emitted due to the decrease in total woody biomass burned.
- The stove provides a safer method of cooking, which in turn help to reduce burn injuries, especially for children, in the household.

c) Economic Sustainability

- The project will help develop a section of the local economy, in the distribution, local assembly, maintenance and monitoring activities.
- Household expenditure on cooking fuel will be reduced through the use of the ICS.
- Saved household labor can be diverted to more productive economic activities.
- The employment opportunity will also equip them with skills which are expected to improve their future employment prospects.

The project is listed under SD VISta[13] and undergoing validation, therefore details of SDGs achieved by the project activity are provided in the SD VISta PD.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable.

[13] https://registry.verra.org/app/projectDetail/SDVISTA/3941

2 SAFEGUARDS

2.1 No Net Harm

The project will only bring positive impacts on environmental and socio-economic aspects as elaborated in Section 1.17. No potential negative environmental or socio-economic impacts have been identified for the project.

2.2 Local Stakeholder Consultation

The overall objective of the consultation process was to disseminate information about the project to stakeholders and obtain their views regarding the project.

The Local stakeholder consultation under the proposed grouped project activity was conducted physically on 17-May-2023 (from 9:30 AM to 12:00 PM South African time) at Hazy view Sun, R40 Road 38km After, White River, Hazyview,1242, South Africa. The local stakeholder meet was organized to gather comments and concerns of the local stakeholders on the proposed grouped project activity. All interested stakeholders representing the following groups: baseline stove users, representatives from local bodies, village heads and implementation partners attended the meeting. The stakeholders were informed by a published notice in the national daily newspapers "Cape times (dated 02-May-2023) & Mpumalanga News (dated 10-May-2023)" Detailed contact information such as phone number and email address of the representative of the project implementer were also made available in the newspaper notice. A total of 33 participants attended the meeting. CQC representatives briefed about the project and the purpose of the stakeholder consultation and then continued with an informative presentation over the project. All the detailed aspects of the project activity were explained including its objectives, technical details of the project stoves, associated social, economic and environmental benefits. The meeting commenced with a welcome note from the CQC Representative, then a presentation of Improved Cook Stoves (ICS) features and benefits by adapting new technology was conducted and followed by a detailed description of the VERRA programme, VCUs, SDVISta and carbon benefits. Also, the PP explained about the significance of the of carbon credits in the project activity and how the carbon finance helps this project feasible. The stakeholders have thanked PP for choosing their region for project implementation. They expressed that the project would lead to sustainable development by reducing deforestation and also improving overall health of the society. After the detailed presentation, stakeholders' interaction was carried out where the stakeholders raised their queries regarding the project activity. The PP has also distributed feedback forms among the participants and collected their feedback.

The meeting documents such as, newspaper notices, email notice, sample feedback forms, etc., are provided in Appendix 1 of this VCS PD.

On-going consultation mechanism

The project proponent will have an on-ground team to collect continuous feedback from end users as they are the most important stakeholders of this project. In addition, the stakeholders can directly contact the project proponent via call or e-mail or letter. These details were shared during the LSC and will also be shared with the end user at the time of household/project stove registration.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

Under the VERRA registry global stakeholder consultation, the project was open for public comments between 31-January-2023 to 02-March-2023.No public comment was received during this period.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

1. VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1[14]
2. Tool 30: "Calculation of the fraction of non-renewable biomass" (version 4.0[15])
3. Standard: "Sampling and Surveys for CDM project activities and programmes of activities" (version 09.0[16])

[14] https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf

[15] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf

[16] https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

3.2 Applicability of Methodology

Applicability criterion	How the project complies
Project activities shall be implemented in domestic premises or in community-based kitchen	The proposed project involves deployment of ICS only in households.
The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	TLC-CQC Rocket stoves planned to be installed under this project are single pot wood cookstoves that have an efficiency of 34.5%. TLC_ rocket Stove efficiency tests[17] performed by Aprovecho Research centre – laboratory for testing the cook stoves.
Both 'Projects' and 'Large Projects' can use this methodology.	The proposed project is expected to generate 1,004,676 tonnes CO_2e per year which is greater than 300,000 tonnes CO_2e per year and thus qualifies as a "large project" as per section 3.10.1 of the VCS Standard version 4.4
Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports, or statistics;	Non-renewable biomass has been used since 31-December-1989 in Republic of South Africa as demonstrated at below.
For the specific case of biomass residues processed as a fuel (e.g., briquettes, wood chips) it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream	Not applicable for the project activity. The ICS is introduced as energy efficiency measure to replace baseline stoves and reduce the use of non-renewable wood fuel (firewood) for cooking.

[17] WBT 4.2.3 efficiency test report provide to VVB as a part of validation.

emissions associated with the processing of the displaced fossil fuel and hence disregarded	
The project design shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	The project proponent will ensure that every cookstove that will be distributed within the project area will have a unique ID comprising of a combination of customer name, geographical location as well as a serial number to identify the ICS end users of this grouped project activity. In addition, the project will be cross checked against other CDM project activity operating in the country using the UNFCCC, the Gold Standard, and other relevant voluntary carbon schemes to ensure that the ICS is not included in any other CDM project activity or voluntary project activity. Also, the approach adopted for distribution of cookstoves and data collection has been elaborately described in sec 5.3 monitoring plan of this VCS PD.
The project design shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	An undertaking will be taken by all the beneficiaries, i.e., the end-users indicating that the ownership of the current project solely lies with C-Quest Capital CR Stoves Pte. Ltd. To further ensure that the ownership rights of the project are reserved with CQC, all the stakeholders relevant, i.e., manufacturers/wholesale providers will provide their consent which will clearly state that the project proponent or an entity authorized by it shall be the sole owner of the VCUs arising from the project.

Evidence that the non-renewable biomass has been in use since 1989

Non-renewable biomass has been in use since 31-December-1989 as evidenced by various FAO statistical data. The Global Forest Resources Assessment 2020 (FAO)[18] indicates that natural forest areas decline yearly. Between years 1990 and 2020 South Africa lost on average of

[18] Global Forest Resources Assessment (FRA) 2020 South Africa - Report
https://www.fao.org/3/cb0074en/cb0074en.pdf (pg 8/56)

36,400 hectares of forest per year. This amounts to an average annual deforestation rate of 0.20% s. In total, between 1990 and 2020, South Africa lost 6.02% of its forest cover, or around 1,092,100 hectares.

Table 4: Decline in forest area between 1990 – 2020

FRA 2020 Categories	Area (1000 hectares)			
	1990	2000	2010	2020
Forest(a)	18,142.09	17,778.09	17,414.09	17,050.09
Other wood land(b)	49,682.43	49,682.43	49,682.43	49,682.43
Other lands (c-a-b)	53,484.48	53,848.48	54,212.48	54,576.48
Total	**121,309**	**121,309**	**121,309**	**121,309**

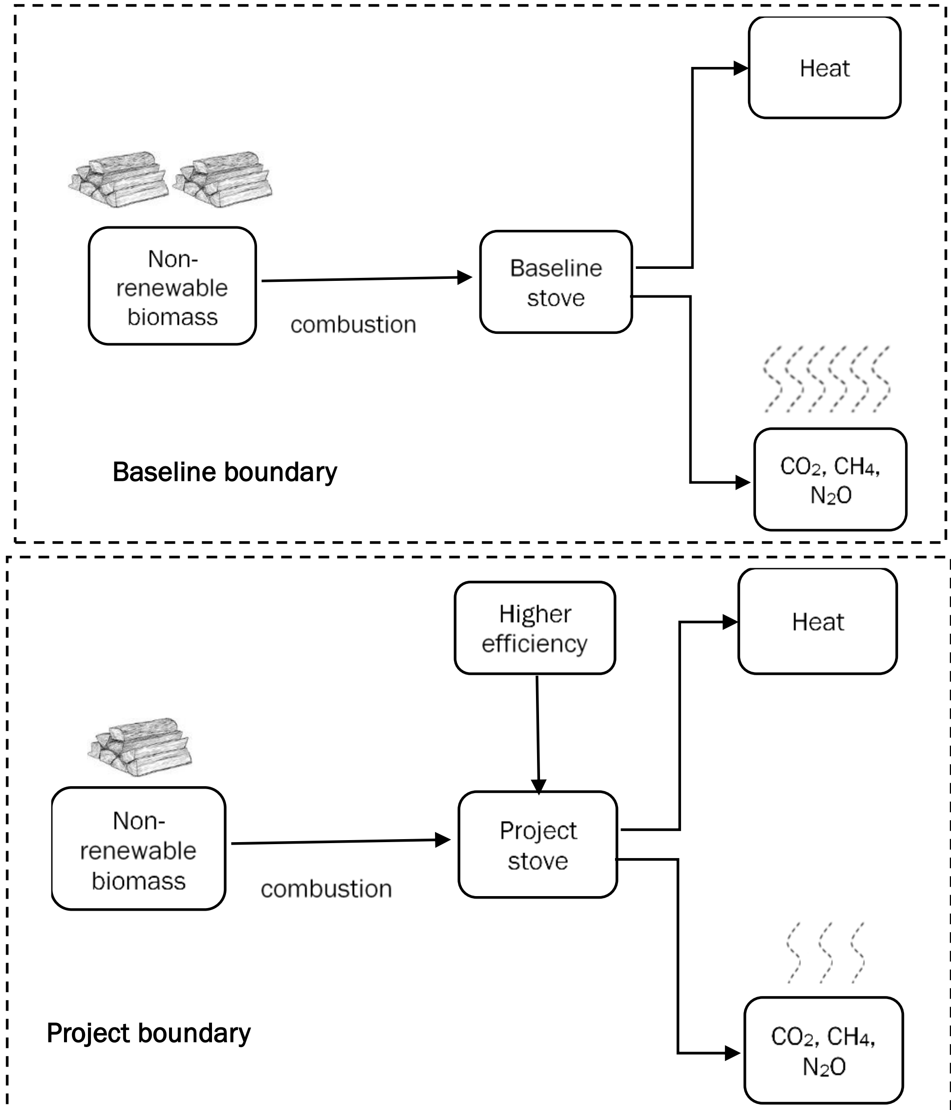
Thus, it can be concluded that losses have been reported in forest areas from before 1989 thereby hinting that fuelwood used across South Africa comes from non-renewable sources.

3.3 Project Boundary

The project location will be the geographical boundary of Republic of South Africa.

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass/Fossil fuel	CO_2	Yes	Major source
		CH_4	Yes	Major source
		N_2O	Yes	Major source
		Other	No	No other source identified
Project	Emission from use of non-renewable biomass	CO_2	Yes	Major source
		CH_4	Yes	Major source
		N_2O	Yes	Major source
		Other	No	No other source identified

A representation of the baseline boundary and project boundary are given as below.



3.4 Baseline Scenario

The baseline scenario is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. The baseline cookstoves are three-stone fires or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system i.e., traditional stoves.

PP carried out a sample baseline scenario assessment in March 2023 to understand the baseline practices and needs of the communities. The baseline scenario assessment evidences the current practice of cooking is using the three-stone stove system using firewood as fuel and with the introduction of ICS, the project will help the target population to reduce consumption of firewood .

However, it is to be noted that LPG and electricity[19] are expensive in the country (South Africa) and are unreliable given the unstable grid situation of the country (South Africa)[20].
Only households with three-stone will be eligible under the grouped project activity and in the absence of group project activity, the target population continues to use traditional three stone fire stove(s) to meet their cooking needs in the baseline. The end users continue to depend on the three stone fire or traditional stoves for cooking i.e., households three stone will be eligible under the grouped project activity. The current project focuses on the replacement of traditional firewood stoves and corresponding savings in the fuel wood for the end users. While the end users might continue to use the other devices, there by the baseline for the current grouped project is replacement of firewood-based cooking in the household by open fire or three stone fired stoves or a conventional device wit*h no improved combustion air supply or flue gas ventilation that is without a grate or a chimney.*
As per the applied methodology, VMR0006 v1.1, used a default value of 10% as efficiency of baseline device i.e., three-stone fire using firewood.

Baseline scenario can also be verified from the end user agreement that will be signed by the registered stove owner at the time of registration of the project device (ICS), which confirms that, registered user voluntarily adopting the project stove and was using the three stone fire or conventional device (without a grate or a chimney) before adopting the project stove.

3.5 Additionality

The methodology uses an activity method for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government program or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by PP.

[19] https://www.news24.com/news24/bi-archive/how-south-africas-electricity-price-compares-to-other-countries-around-the-world-2022-2
[20] https://www.datacenterdynamics.com/en/opinions/the-effects-of-a-failing-power-grid-in-south-africa/
https://energy.economictimes.indiatimes.com/news/power/south-africa-sees-worst-power-cuts-on-record-in-2020-research-shows/77821620

Step 2: Positive List

The applicability conditions of this methodology represent the positive list. The project must demonstrate that it meets all of the applicability conditions of the methodology as well as the below condition. In so doing, the project is deemed as complying with the positive list.

1. Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.

2. Project activities that are implemented as part of government schemes or are supported by multilateral funds cannot be considered additional even if the stoves are distributed free of cost or at a highly subsidized rate and hence are not eligible to use this methodology.

As per Section 3.2, the project meets the applicability conditions of the methodology which represent the positive list.

The project installs the ICS at zero cost to the household and has no other source of revenue other than the sale of GHG credits.

The project is not implemented as part of government schemes or supported by multilateral funds.

Conclusion: As the project fulfills the conditions above, it is deemed additional.

Step 3: Project Method

For any project activity where stoves are not provided at zero cost to the end-user or has any other source of revenues other than the sale of GHG credits, then the project activity shall apply investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality included in AMS-II.G to determine that the proposed project activity is either: 1) not the most economically or financially attractive, or 2) not economically or financially feasible

This step is not applicable under project activity because the stoves are provided at zero cost to the end-users.

3.6 Methodology Deviations

The project did not apply any methodology deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The applied methodology VMR0006 Vesrion1.1 does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4

4.2 Project Emissions

The VMR0006 Version1.1 methodology does not account for project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with applied methodology VMR0006, Version1.1.

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the applied methodology will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \qquad \text{Equation (1)}$$

Where:

i	=	Indices for the situation where more than one type/model of improved cookstove is introduced to replace three-stone fire
j	=	Indices for the situation where there is more than one batch of improved cookstove of type *i*
ER_y	=	Emission reductions during year *y* in t CO_2e
$ER_{y,i,j}$	=	Emission reductions by improved cookstove of type *i* and batch *j* during year *y* in t CO_2e

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times \left(EF_{wf,CO2} + EF_{wf,non\ CO2}\right) \times N_{y,i,j} \times 0.95 \qquad \text{Equation (2)}$$

Where:

$B_{y,savings,i,j}$	=	Quantity of woody biomass that is saved in tonnes per improved cookstove of type *i* and batch *j* during year *y*
$f_{NRB,y}$	=	Fraction of woody biomass that can be established as non-renewable biomass (fNRB)[21]
$NCV_{wood\ fuel}$	=	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)[22]
$EF_{wf,CO2}$	=	CO_2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO_2/TJ)[23]
$EF_{wf,non\ CO2}$	=	Non-CO_2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO_2/TJ)[24]
$N_{y,i,j}$	=	Number of improved cookstoves of type *i* and batch *j* operating during year *y*
0.95	=	Discount factor to account for leakage

The quantify of woody biomass saved due to implementation of improved cookstoves to be estimated using equation below:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1\right) \qquad \text{Equation (3)}^{25}$$

where

η_{old}	=	Efficiency of baseline cookstove
$\eta_{new,y,i,j}$	=	Efficiency of the improved cookstove type *i* and batch *j* determined through water boiling test (WBT) during year *y* Alternatively, efficiency may be determined using Equation (4)[26].
$B_{y=1,new,i,j,survey}$	=	Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type *i* and batch *j*, determined in the first year of the implementation of the project through a sample survey.

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94 \qquad \text{Equation (4)}$$

[21] f_{NRB} Value for the Project region i.e., Republic of South Africa by third party as per CDM TOOL 30 Version 04.

[22] 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction

[23] 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

[24] 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

[25] Equation (3) in PD is Equation (4) as per applied methodology VMR0006, V1.1

[26] Equation (4) in PD is Equation (5) as per applied methodology VMR0006, V1.1

where

η_p	=	Efficiency of project stove (fraction) at the start of project activity.
$(DF_n)^{y-1}$	=	Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.
0.94	=	Adjustment factor to account for uncertainty related to project cookstove efficiency test.

The f_{NRB} has been calculated according to the provision of the VCS methodology VMR0006, Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1. VMR0006 requires the use of CDM's TOOL30 "Calculation of the fraction of non-renewable biomass" Version 4.0.

To calculate f_{NRB} following the equations were used and as follows.

$$fNRB = \frac{NRB}{NRB + RB}$$ Equation (5)[27]

$$NRB = H - RB$$ Equation (6)[28]

$$H = HW\ X\ N + C + NE$$ Equation (7)[29]

$$RB = \sum(MAI\ forest, i\ X\ (F\ forest, i - P\ forest, i)) + \sum(MAI\ other, i\ X\ (F\ other, i - P\ other, i))$$ Equation (8)[30]

Where,

NRB	=	Quantity of non-renewable woody biomass(tonnes)
RB	=	Quantity of renewable woody biomass production(tonnes)
H	=	Total annual wood consumption (energy/non energy) in the country(tonnes)

[27] As per Equation (1) of the applied Tool 30 "Calculation of fraction of non-renewable biomass ", Version 04
[28] As per Equation (2) of the applied Tool 30 "Calculation of fraction of non-renewable biomass ", Version 04
[29] As per Equation (3) of the Tool 30 "Calculation of fraction of non-renewable biomass ", Version 04
[30] As per Equation (4) of the Tool 30 "Calculation of fraction of non-renewable biomass ", Version 04

HW = Average HH wood fuel (including charcoal) consumption[31] (tonnes/household)

N = Number of HHs consuming wood fuel in the region/project area.(number)

CE = Commercial woody biomass consumption for energy application.[32] (tonnes)

NE = Commercial woody biomass consumption for non- energy application(tonnes)

MAI = Mean annual increment in forest/other wooded land[33] (tonnes/ha/yr.)

F = Extent of forest/other wooded land[34] (ha)

P = Extent of non -accessible area (protected area within forest/other wooded land.)[35] (ha)

TOOL30 Version 04 provides for the calculation of the project specific f_{NRB} on a regional basis, which includes the utilisation of recent, national, and local government statistics. The quantity of NRB has therefore been determined by calculating the total consumption of wood (H) in the Republic of South Africa and then deducting the quantity of renewable biomass (RB) from this value[36]. The results are as follows for f_{NRB} calculations.

Table 5: F_{NRB} Calculation[37]

Variable	Unit	Value
Total woody biomass consumption (H)	t/yr.	41,459,054
Renewable biomass (RB)	t/yr.	5,409,740
Non-renewable biomass (NRB)	t/yr.	36,049,313[38]
Fraction of non-renewable biomass (f_{NRB})	-	**0.87**[39]

Determination of number of ICS operating during year y

$N_{y,i,j}$ = 500,000 x [1 - (y-1) x 10%]

Example of calculation:

[31] Approved standardized baseline or Published literature/research paper –Government data, organizations of repute (FAO, UN etc.) or Survey results or Default value as per Tool 30
[32] Energy Statistics Database -UN Statistics Division; Existing studies; Government data; Survey.
[33] IPCC Good Practice Guidelines, Chapter 4 Table 4.9, www.ipcc-nggip.iges.or.jp or Global Forest Resources Assessment www.fao.org or National studies or government data or official statistics.
[34] Global Forest Resources Assessment by the Food and Agriculture Organization of the United Nations (FAO); Official statistics: Project specific survey data.
[35] Global Forest Resources Assessment by the Food and Agriculture Organization of the United Nations (FAO); National studies or Government data or Official statistics
[36] The calculations spread sheet of f_{NRB} provided to VVB.
[37] This parameter determined ex-ante value. PP has contracted an independent party "C4Ecosolutions" for calculation of f_{NRB} as per CDM Methodological Too 30l: "Calculation of fraction of non- renewable biomass" (v04.0) for Republic of South Africa. f_{NRB} calculation sheet provided to VVB as a part of validation.
[38] total woody biomass (H) minus renewable biomass (RB)
[39] quotient of non-renewable biomass (NRB) and the sum of RB and NRB

If y = 2,

$N_{y,i,j}$ = 500,000 x [1 - (2-1) x 10%]

= 500,000 x 90%

= 450,000

Hence, the number of ICS operating during year y is as below:

Year (y)	$N_{y,i,j}$
01-February-2024 to 31-January-2025	500,000
01-February-2025 to 31-January-2026	450,000
01-February-2026 to 31-January 2027	400,000
01-February-2027 to 31-January-2028	350,000
01-February-2028 to 31-January-2029	300,000
01-February-2029 to 31-January-2030	250,000
01-February-2030 to 31-January-2031	200,000
01-February-2031 to 31-January-2032	150,000
01-February-2032 to 31-January-2033	100,000
01-February-2033 to 31-January-2034	50,000

Determination of efficiency of ICS during year y

$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$

Where

η_p = 34.5%

DF_n = 0.99

Example of calculation:

If y= 2

$\eta_{new,y,i,j}$ = 34.5% X $(0.99)^{2-1}$ X 0.94

= 32.11%

Hence the efficiency of ICS during year y is as below:

Year (y)	$\eta_{new,y,i,j}$
01-February-2024 to 31-January-2025	32.43%
01-February-2025 to 31-January-2026	32.11%
01-February-2026 to 31-January 2027	31.78%
01-February-2027 to 31-January-2028	31.47%
01-February-2028 to 31-January-2029	31.15%
01-February-2029 to 31-January-2030	30.84%
01-February-2030 to 31-January-2031	30.53%
01-February-2031 to 31-January-2032	30.23%
01-February-2032 to 31-January-2033	29.92%
01-February-2033 to 31-January-2034	29.63%

Determination of quantity of woody biomass that is saved in tonnes per ICS during year y

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1\right)$$

Example of calculation:

If y= 2,

$B_{y,savings,i,j}$= 0.954 X [(0.3211/0.1) -1]

= 2.1099 tonnes

Year (y)	$B_{y=1,new,i,survey}$	$\eta_{new,y,i,j}$	η_{old}	$B_{y,savings,i,j}$
01-February-2024 to 31-January-2025	0.954	32.43%	10%	2.1409
01-February-2025 to 31-January-2026	0.954	32.11%	10%	2.1099
01-February-2026 to 31-January 2027	0.954	31.78%	10%	2.0793
01-February-2027 to 31-January-2028	0.954	31.47%	10%	2.0490

01-February-2028 to 31-January-2029	0.954	31.15%	10%	2.0189
01-February-2029 to 31-January-2030	0.954	30.84%	10%	1.9892
01-February-2030 to 31-January-2031	0.954	30.53%	10%	1.9597
01-February-2031 to 31-January-2032	0.954	30.23%	10%	1.9306
01-February-2032 to 31-January-2033	0.954	29.92%	10%	1.9018
01-February-2033 to 31-January-2034	0.954	29.63%	10%	1.8732

Determination of emission reductions by ICS of batch 1 during year y

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO2} + EF_{wf,non\ CO2}) \times N_{y,i,j} \times 0.95$$

Where

$NCV_{wood\ fuel}$ = 0.0156 TJ/tonne

$f_{NRB,y}$ = 0.87

$EF_{wf,CO2} + EF_{wf,non\ CO2}$ = 112 + 26.23 = 138.23 tCO_2/TJ

Example of calculation:

If y=2,

$ER_{y,i,j}$ = 2.1099 X 0.0156 X 0.87 x 138.23 X 450,000 X 0.95

=1,692,197 tCO_2

Year	$B_{y,savings,i,j}$	$NCV_{wood\ fuel}$	$f_{NRB,y}$	$EF_{wf,CO2} + EF_{wf,non\ CO2}$	$N_{y,i,j}$	$ER_{y,i,j}$
01-February-2024 to 31-January-2025	2.1409	0.0156	0.87	138.23	500,000	**1,907,803**
01-February-2025 to 31-January-2026	2.1099	0.0156	0.87	138.23	450,000	**1,692,197**
01-February-2026 to 31-January 2027	2.0793	0.0156	0.87	138.23	400,000	**1,482,329**
01-February-2027 to 31-January-2028	2.0490	0.0156	0.87	138.23	350,000	**1,278,114**
01-February-2028 to	2.0189	0.0156	0.87	138.23	300,000	**1,079,467**

31-January-2029						
01-February-2029 to 31-January-2030	1.9892	0.0156	0.87	138.23	250,000	**886,308**
01-February-2030 to 31-January-2031	1.9597	0.0156	0.87	138.23	200,000	**698,553**
01-February-2031 to 31-January-2032	1.9306	0.0156	0.87	138.23	150,000	**516,124**
01-February-2032 to 31-January-2033	1.9018	0.0156	0.87	138.23	100,000	**338,941**
01-February-2033 to 31-January-2034	1.8732	0.0156	0.87	138.23	50,000	**166,925**

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
01-February-2024 to 31-January-2025	-	-	-	1,907,803
01-February-2025 to 31-January-2026	-	-	-	1,692,197
01-February-2026 to 31-January 2027	-	-	-	1,482,329
01-February-2027 to 31-January-2028	-	-	-	1,278,114
01-February-2028 to 31-January-2029	-	-	-	1,079,467
01-February-2029 to 31-January-2030	-	-	-	886,308
01-February-2030 to 31-January-2031	-	-	-	698,553
01-February-2031 to 31-January-2032	-	-	-	516,124
01-February-2032 to 31-January-2033	-	-	-	338,941
01-February-2033 to 31-January-2034	-	-	-	166,925
Total	-	-	-	**10,046,761**

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$f_{NRB,y}$

Data unit	Fraction
Description	Fraction of wood that can be established as non-renewable biomass
Source of data	Official statistics or reports or peer-reviewed literature (IPCC, UN Data & FAO reports)
Value applied	0.87
Justification of choice of data or description of measurement methods and procedures applied	This parameter determined ex-ante. C4 EcoSolutions (Pty) Ltd was appointed as third party to study and derive the f_{NRB} value for South Africa; using most recent historical data available from Food and Agriculture Organization (FAO) report, UN Data and other publicly available data that have been published by reliable sources. As per applied CDM Tool 30, Version 04 value is calculated as per following equation of the applied tool[40], F_{NRB} = NRB/(NRB+RB) f_{NRB} = Fraction of non-renewable biomass (fraction or %) NRB = Quantity of non-renewable biomass (t/yr.) RB = Quantity of renewable biomass (t/yr.) Calculation provided in sec 4.4 of this VCS PD.
Purpose of Data	Calculation of emission reductions
Comments	f_{NRB} calculations sheet provided to VVB during the validation.

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction[41]
Value applied	0.0156
Justification of choice of data or description of	IPCC default value

[40] Equation (1) of the applied Tool 30 Version 04.

[41] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_1_Ch1_Introduction.pdf (page 19/29)

measurement methods and procedures applied	
Purpose of Data	Calculation of emission reductions
Comments	None

Data / Parameter	$EF_{wf,CO2}$
Data unit	tCO_2/TJ
Description	CO_2 emission factor for the use of wood fuel in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	None

Data / Parameter	$EF_{wf,non\ CO2}$
Data unit	tCO_2/TJ
Description	Non-CO_2 emission factor for the use of wood fuel in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	None

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity.
Source of data	.Stove efficiency test as per WBT 4.2.3[42] by Aprovecho Research centre – laboratory for testing the cook stove
Value applied	0.345
Justification of choice of data or description of measurement methods and procedures applied	This parameter determined ex-ante
Purpose of Data	Calculation of $\eta_{new,y,i,j}$
Comments	None

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type I and batch j operating during year y
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities.
Frequency of monitoring/recording	At least once every two years
Value applied	For ex-ante emission reduction calculation, it is assumed that the project will distribute up to 500,000 ICS.
Monitoring equipment	Monitoring survey

[42] Testing protocol WBT 4.2.3 has been followed

QA/QC procedures to be applied	Sampling standard “sampling and surveys for CDM project activities and programme of activities” version 9 will be used for determining the sample size to achieve 90/10 confidence precision.
Purpose of data	Calculation of emission reductions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value
Comments	None

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline stove (for the three stone fire system with firewood)
Source of data	Methodological default value
Description of measurement methods and procedures to be applied	A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.
Frequency of monitoring/recording	Fixed for each individual household at the time of project implementation.
Value applied	0.1
Monitoring equipment	Not Applicable
QA/QC procedures to be applied	Baseline stove use can be confirmed from the registration form cum consent deed signed by each Household/end user at the time of installation of ICS which states that the “household does not currently own an improved cookstove”.
Purpose of data	Calculation of emission reductions
Calculation method	Not Applicable
Comments	Default value of efficiency 10% or 0.1 considered as per applied methodology VMR0006, V1.1

Data / Parameter	$\eta_{new,y,i,j}$

Data unit	Fraction
Description	Efficiency of the improved cookstove type *i* and batch *j* determined through water boiling test (WBT) during year *y*
Source of data	Calculation
Description of measurement methods and procedures to be applied	To adopt Option V given in the applied methodology parameter table: "Efficiency of the improved cookstoves to be estimated using methodology equation 5 above[43] where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored"
Frequency of monitoring/recording	Annually
Value applied	For ex-ante calculation, the value below is applied. <table><tr><th>Year (y)</th><th>$\eta_{new,y,i,j}$</th></tr><tr><td>1</td><td>32.43%</td></tr><tr><td>2</td><td>32.11%</td></tr><tr><td>3</td><td>31.78%</td></tr><tr><td>4</td><td>31.47%</td></tr><tr><td>5</td><td>31.15%</td></tr><tr><td>6</td><td>30.84%</td></tr><tr><td>7</td><td>30.53%</td></tr><tr><td>8</td><td>30.23%</td></tr><tr><td>9</td><td>29.92%</td></tr><tr><td>10</td><td>29.63%</td></tr></table>
Monitoring equipment	Calculated value (No Monitoring equipment used)
QA/QC procedures to be applied	This parameter will be calculated using equation (5)[44] of the applied methodology.
Purpose of data	Calculation of emission reductions

[43] Equation (5) of applied methodology VMR0006 V1.1 is Equation (4) in the VCS PD

[44] Equation (5) of applied methodology VMR0006 V1.1 is Equation (4) in the VCS PD

Calculation method	Calculation to be performed using equation below: $\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$
Comments	None

Data / Parameter	$\boldsymbol{B_{y=1,new,i,j,survey}}$
Data unit	Tonnes
Description	Quantity of woody biomass used by project devices in tonnes per device of type i
Source of data	Monitoring survey (In the first year of introduction of ICS)
Description of measurement methods and procedures to be applied	Minimum sample size of each type i and batch j should be in line with the latest version of Standard for sampling and surveys for CDM project activities and programme of activities[45]. Determined in the first year of the introduction of the devices (e.g. during the first year of the crediting period, y=1) through measurement campaigns at representative households and/or sample survey. Sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (i.e. that do not implement measurement campaigns) may only be used if the following conditions are satisfied. (a) Baseline cookstoves have been completely decommissioned and only improved cookstoves are exclusively used in the project households; (b) If multiple devices are used in the project, it is possible from the results of the survey questions to clearly differentiate the quantity of firewood being used by each device. In other words, if more than one device, or another device that consumes firewood, are in use in project households, then the sample survey needs to distinguish the quantity of firewood used by the project device and the other devices that use firewood. $B_{y=1,new,I,j,survey}$ parameter determined in the first year of the introduction of the devices through measurement campaigns at representative households. Under the grouped project activity two TLC-CQC Rocket Stoves will be installed in each of the target households, which are termed project stove 01 and project stove 02. At the time of

[45] *Standard for Sampling and Surveys for CDM Project Activities and Program of Activities, version 09.0*.

	monitoring survey, field staff asks the end user/beneficiary HH to make a pile for the total firewood required for cooking in a day for all the stoves available in his/her house. Further user was asked to extract and make the piles for the wood required for the project stove 1 and project stove 2 separately from that pile and weigh both the piles. Same has been recorded in the survey forms and in the spreadsheet. Therefore, firewood consumed for each project stove(s) and baseline stove if any being used can be distinguished clearly. Detailed description provided in Sec 5.3 under item field measurements.
Frequency of monitoring/recording	Determined in the first year of project implementation
Value applied	For ex-ante calculation, the value is assumed as 2.615 kg/device/day[46] or equal to 0.954 tonnes/device/year.
Monitoring equipment	Measurements undertaken using weighing Scales for quantity of fuel wood consumed by project stoves.
QA/QC procedures to be applied	Survey to be conducted as per requirements of the methodology. Monitoring equipment if any will be calibrated as per the requirements/manufacturer 's guidelines.
Purpose of data	Calculation of emission reductions
Calculation method	Mean woody biomass consumption of sample stoves to be applied across entire population using a particular model of stove.
Comments	Value determined by monitoring survey conducted during first verification.

Data / Parameter	**Life Span**
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation (5)[47] is adopted to determine the project stove efficiency

[46] For ex-ante estimates value assumed based on the registered CDM PoA 10443 in the project region (https://cdm.unfccc.int/ProgrammeOfActivities/poa_db/KUFLHQD3N8E5W2BIMRAVTGJXO14S76/viewCPAs

Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	TLC-CQC rocket cookstoves are built on-site at the end user premises as per the design specification. This will be achieved using brick molds of specified dimensions to make bricks locally that are suitable for the stove construction . PP provides the metal parts to implementing partner /end user for the installation and registration of ICS . PP will conduct periodic audits and visits the stoves to ensure effective usage throughout the project lifetime. This along with spot audits and maintenance services, will ensure that the project stoves will continue to work efficiently. TLC stove a clay and brick stove which can be easily maintained & repaired using locally available materials. The metal parts provided by the pp are specifically design for the project stove using high quality material for reliability & durability for project life of 10years. Any damages, worn out metal parts will be replaced and provide by PP for the project lifetime for their continuous usage. Hence the stoves will be functioning for a period of 10 years.
Frequency of monitoring/recording	Once at the time of project stove installation
Value applied	10
Monitoring equipment	No equipment is required to monitor this parameter
QA/QC procedures to be applied	Not applicable
Purpose of data	Calculation of emission reductions
Calculation method	No calculation used for this parameter
Comments	None

Data / Parameter	MC_y
Data unit	Count (Number)

[47] Equation (5) of applied methodology VMR0006 V1.1 is Equation (4) in the VCS PD

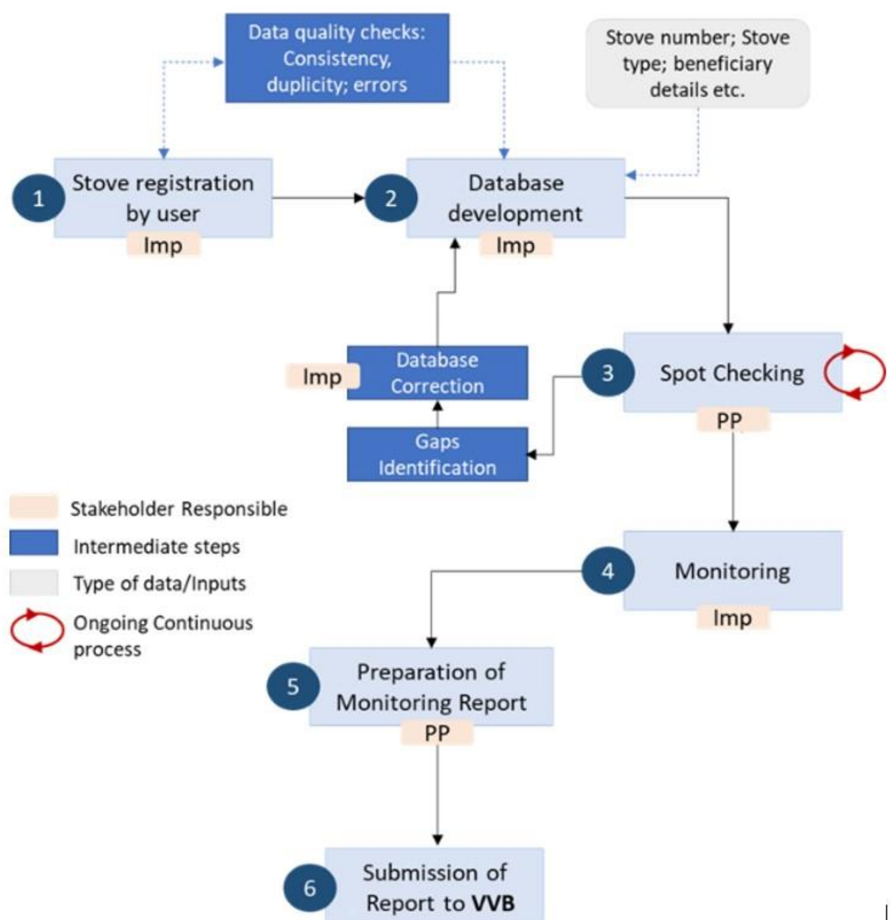
Description	Number of meals cooked on project stoves by the households per week (meal count)
Source of data	Monitoring survey captures the data based on a survey questionnaire
Description of measurement methods and procedures to be applied	As described in the baseline scenario, there is a probability of multiple stoves usage in some of the project households (HHs), along with the project stoves. In such a scenario, it becomes critical to distinguish the exclusive project stove contribution to the meals prepared by the HHs to appropriately calculate the emission reductions achieved by the project. To ensure this, PP will adopt a weekly meal count approach, along with the project stove fuel consumption measurements to arrive at realistic emission reductions. This approach allows monitoring of project and other stoves contributions in the kitchen and claim for only the eligible portion by the project stoves. The survey questionnaire for the sample HHs will have exclusive sections to record the total number of meals cooked by the HH per week and the number of meals contributed by each project stove to the total meals. The suveys will be conducted in conjunction with the $B_{y=1,new,ij,survey}$ parameter defined above, for the same sample households selected ruing the first monitoring period.
Frequency of monitoring/recording	Once at the time of project stove installation
Value applied	For exnate, it is assumed that all the meals are cooked on the project stoves. Acutal values determined expost.
Monitoring equipment	No equipment is required to monitor this parameter. Surveys will be administered through a detailed questionnaire.
QA/QC procedures to be applied	Survey to be conducted as per the requirements of the methodology. Best practices will be adopted in collection of the data through questionnaires inline with the accepted research practices.
Purpose of data	Calculation of Fuel wood consumption
Calculation method	Detailed calculation method is provided in section 5.3 under 'Share of project devices (Meal count based approach)'.
Comments	None

5.3 Monitoring Plan

The Monitoring Plan applied in this project involves several key elements that ensure that the PP and/or project Implementer have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating Verified Carbon Units (VCUs) following VCS methodology (VMR0006 version 1.1) on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

- Data collection procedures
- Distribution and Monitoring Database
- Spot Checking of ICS (ongoing)
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

The below flow-chart illustrates the roles and responsibilities of the parties during the implementation of the monitoring plan for the project. In the below flowchart, the project implementer is abbreviated to "Imp" and can be the PP or another party authorized by the PP



<u>Roles and Responsibilities of stakeholders during the implementation of the monitoring plan as per above steps on the flow chart</u>

1. **Implementor: User registers stove:** Project implementer will collect/receive the necessary information requested in the Registration process from the user. Means of collecting this information may be through a physical registration card filled by project Implementor staff, retailers, end-users, or partner organization's staff, or using ICTs or SMS. Project Implementers' staff shall double check the accuracy of the information provided, and request for field staff additional clarifications if needed.

2. **Implementor: Data logged into database:** Project implementer's trained staff will input the data in the database either manually (if data collected from physical registration card) or this will be automatically input if data was collected using ICTs or SMS. Project implementer staff shall double check the information included on the database and check for duplications. Any duplicate information shall be investigated, and errors corrected or excluded from the database if it is a true duplicate entry.

3. **Project Proponent: Spot- checking (ongoing):** Project Proponent has a procedure for internal auditing called spot-checking, where field staff randomly selects households to ensure proper functioning of the project stoves throughout the project lifetime and to ensure that the project stoves continue to work as efficient as specified by the manufacturer. Field staff of project proponent randomly select units included in the database and visit or contact the stove owners to cross-check the information on the database with the factual evidence in the field. Any inconsistencies found (e.g., change in the address of a user) get updated on the database, and in case, ICS are found to be no longer in use, they get clearly marked as such and is excluded from emission reductions calculations.

   During the spot check, if any of the project stoves was found to be in poor working condition like damaged or in unusable condition, the stove owner may repair the stove for minor issues. If the stove owners are unsuccessful in repairing or solving the issues on their own, the owner may reach out to CQC & IP through helpline, field personnel or the village headmen for further assistance. Respective channel representative, either headman, field staff, IP, helpline executive or authorized personnel informs CQC or their officials for appropriate action and to resolve the grievance raised by the beneficiary. CQC or their representatives will keep a record of such incidents and ensure the resolution is achieved to the satisfaction of the end users.

4. **Implementor: Monitoring:** Project implementer will follow the requirements as per PD to collect the necessary information for a monitoring report. The project proponent will assess all monitoring data and produce a monitoring report for the VVB to verify corresponding to the preceding monitoring period. This report will present the data relating to the emission reductions generated by those batches during the monitoring period.

5. **Project Proponent: Preparation of monitoring report:** the project implementers or the PP will prepare the final monitoring report to be provided to the VVB for verification of emission reductions. A copy of the monitoring report will remain with the Project Proponent.

   The PP will coordinate and manage each project Implementer and assist them in implementing each element of the monitoring plan. The monitoring plan shall be elaborated in accordance with the Sampling Plan below. Also, regular audits conducted throughout the project lifetime ensures proper functioning of stoves and determination of the lifespan of the same.

Procedures for stove construction, sourcing, and procurement.

TLC-CQC rocket cookstoves are built on-site at the end user premises as per the design specification.

This will be achieved using brick molds of specified dimensions to make bricks locally that are suitable for stove construction. PP provides the metal parts to implementing partner/end user for the installation and registration of ICS .

PP ensures the IP/ households and village members are aware & appropriately trained to construct, maintain, and repair the project stoves. End users are provided with information/details to communicate with IP/PP in case of further assistance/support required in repairing & maintenance of project stoves.

PP will conduct periodic audits and visit the stoves to ensure effective usage throughout the project lifetime. This, along with spot audits and maintenance services, will ensure that the project stoves will continue to work efficiently.

TLC stove a clay and brick stove which can be easily maintained & repaired using locally available materials. The metal parts provided by the pp are specifically designed for the project stove using high quality material for reliability & durability for project life of 10years. Any damage, worn out metal parts will be replaced and provide by PP for the project lifetime for their continuous usage. Hence the stoves will be functioning for a period of 10 years as provide in VCS PD under the parameter table life span in Sec 5.2

Sampling Plan

As per the *Guideline for Sampling and Surveys for CDM Project Activities and Program of Activities*, *version 04.0*, the sampling plan is the following:

(a) Sampling design

Due to the large number of ICS envisioned to be distributed under the project, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling shall be undertaken for ICS. The sampling plan to be designed in line with the requirements of the *Standard for Sampling and Surveys for CDM Project Activities and Program of Activities, version 09.0*.

Samples will be drawn from ICS recorded in the project database administered by the project proponent. A detailed explanation of this database is found in Section 1.11 of the project description.

i. Objectives and Reliability Requirements

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period and with 90/10 confidence/precision (as per applied methodology VMR0006; version 1.1).

Monitored Parameters:

The project involves introduction of improved cook stoves as energy efficiency measure:

No.	Monitoring parameters	Sampling parameters	Parameter type	Monitoring frequency
1	$N_{y,i,j}$ Number of project	Proportion of ICS still in operation	Proportion	At least once in two years (Biennially)

i *i*	devices of type *i* and batch *j* operating during year *y*			
. 2 *Target*	$B_{y=1,new,i,j,survey}$ Quantity of woody biomass used by project devices in tonnes per device of type *i*. and batch j	Daily consumption of woody biomass per ICS	Mean value	Determined in the first year of project implementation

Target Population

The target population will be the complete set of appliances (ICS) deployed under the project.

iii.Sampling Frame

It is expected that the geographical locations do not have influence on the parameters of interest that is $N_{y,i,j}$, & $B_{y=1,new,i,survey}$., Therefore, these monitoring parameters can be assumed to be highly homogeneous how the end user group and distribution/installation location is defined in the following paragraphs.

The sample frame refers to all the information sources on the Database. The SMS/ICT system is the primary mechanism for data collection for newly distributed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the project. The SMS/ICT data is used to populate the stoves Database and the Monitoring Survey follows the "Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities"; version 09.

All the project stoves distributed under this project will be grouped together to create a Primary Sampling Unit which is homogenous. The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using, a 90 per cent confidence level, and a 10 per cent margin of error, a random sample will be selected from the target population.

Share of project devices (Meal count based approach) .

As described in the baseline scenario, there is a probability of multiple stoves usage in some of the project households (HHs), along with the project stoves. In such a scenario, it becomes critical to distinguish the exclusive project stove contribution to the meals prepared by the HHs to appropriately calculate the emission reductions achieved by the project. To ensure this, PP has adopted a weekly meal count approach, along with the project stove fuel consumption measurements to arrive at realistic emission reductions.

This approach allows monitoring of project and other stoves contributions in the kitchen and claim for only the eligible portion by the project stoves. The survey questionnaire for the sample

HHs will have exclusive sections to record the total number of meals cooked by the HH per week and the number of meals contributed by each project stove to the total meals. To avoid biases and uncertainties, PP will monitor contributions of all types of stoves in the kitchen that shall match 100% of the meals prepared by the HH per week.

On the other hand, per day fuel wood consumption measurement of the household and along with the number of exclusive meals prepared per day using the measured fuel wood is monitored as part of the survey. This will help arrive at the average per meal fuel wood consumption (considered as a unit), which will then be used to multiply with number of weekly meals prepared by the HH using project stoves, resulting in the total fuel wood consumption of the HH per week for each project stove. This will be further multiplied with the exact number of weeks in the respective monitoring period or year to arrive at the total fuel consumption.

By determining the average number of meals prepared on the project stoves (along with other operating stoves), PP ensures that the project stove replaces only the baseline traditional stove which otherwise would have been used in the absence of the project stoves/ICS.

A Sample calculation is provided below for reference.

Unit fuel consumption (kgs of fuel/meal) =

Total wood fuel consumed per day by project stove (kg)/number of meals cooked on project stove per day.

Fuel consumption of project stove 1 (kgs/week) = Number of meals prepared on project stove 1(week) X Unit fuel consumption (kgs of fuel/meal)

Total fuel consumption by project stove 1 (tonnes/monitoring period) = (Fuel consumption of project stove (kgs/week) X Number of weeks in the monitoring period) /1000

<u>Reference calculations:</u>

Assuming -

Total wood fuel consumed per day by project stove (kg) = 5 kg/day

Number of meals cooked on project stove per day = 3/day

Unit fuel consumption (kgs of fuel/meal) = 5/3 = 1.67 kg/meal

Assuming -

Number of meals prepared on project stove 1 (week) = 18

Unit fuel consumption (kgs of fuel/meal) = 1.67 kg/meal (as above)

Fuel consumption of project stove 1 (kgs/week) = 18 X 1.67 =30.06 kgs/week

Assuming –

Fuel consumption of project stove (kgs/week) = 30.06 kg (as above)

Number of weeks in the monitoring period = 24 weeks

Total fuel consumption by project stove 1 (tonnes/monitoring period) =

30.06 X 24 /1000 = 0.72 tonnes per MP (six months) or 1.44 per annum

The same formulas and calculation method described above will be applied for project stove 2 to arrive at its respective fuel consumption as desired.

The number of samples for collection of this data will be determined as per the sampling frame elaborated below.

1) Sampling frame for proportion of ICS still in operation ($N_{y,j}$)

The first step is to identify the Primary Sampling Units.

Factors such as model of the ICS do not effect parameter $N_{y,j}$. Therefore Primary sampling unit in case of parameters $N_{y,j}$ is independent of model of stove.

2) Quantity of woody biomass used by project devices in tonnes per device ($B_{y=1,new,i,\ survey}$)

The parameter $B_{y=1,new,i,\ survey}$ shall vary in accordance with its model. Hence for this parameter, the Primary Sampling Unit shall be defined as the group of ICSs of the same model. If two different ICS models being implemented in the same project – this will form two Primary Sampling Units. The value will be determined in the first year of the introduction of the devices through the monitoring survey.

Under this project up to two project stoves will be installed in each household, which are classified as Project stove 1 and project stove 2. At the time of monitoring survey, field staff record the same in survey app. Therefore, firewood consumed for each project stove can be distinguished clearly. For measuring the fuel wood, calibrated weighing scales will be used to ensure data quality and accuracy.

iv. Sampling Method

The sampling method for the monitored parameters $N_{y,i,j}$, and $B_{y=1,new,i,survey}$ is Simple Random Sampling and samples will be randomly selected from the primary sampling units as illustrated above. To ensure a random selection of ICS, random number generators shall be applied.

Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$\boldsymbol{N_{y,i,j}}$	Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)
$\boldsymbol{B_{y=1,new,i,j,survey}}$	Interview with end user and estimate the daily consumption of woody biomass of ICS (First year of implementation)

v. Sample Size

For the estimation of the proportion or mean value of the parameters investigated, the minimum sample size for each sample frame has to achieve the 90/10 confidence/precision as defined in sampling design above,.

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using, a 90 per cent confidence level, and a 10 per cent margin of error, a random sample will be selected from primary sampling .

The parameter number of stoves still in operation during the monitoring period as determined by the monitoring survey (Ny,j) will be estimated through sampling annually/biennially and the parameter Quantity of woody biomass used by project devices in tonnes per device ($B_{y=1,new,i, survey}$) will be estimated through sampling within the first year of project implementation . $B_{y=1,new,i, survey}$ can be coupled with $N_{y,j,j}$ for the first year. Of the two parameters to be monitored, $N_{y,i,j}$ is proportion/percentage parameter and $B_{y=1,new,i, survey}$ is a mean value parameter.

In order to calculate the required sample size estimates, values for the proportions, mean values, and standard deviations are required. As per *Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 04.0*, there are different ways available to obtain the estimates of the parameter of interest:

(a) Refer to the result of previous studies and use these results;

(b) In a situation where information from previous studies is not available, a preliminary sample as a pilot could be conducted and use that sample is used to provide the estimates;

(c) Use best guesses based on the researcher's own experiences.

(i) Proportion parameter ($N_{y,i,j}$)

To estimate the sample size for proportion parameters, the following equation[48] is used:

$$n \geq \frac{1.645^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 \times p(1-p)}$$

Where:

n	= Sample size
N	= Population size (Total number of households/ICS)
p	= Expected proportion
1.645	= Represents the 90% confidence required
0.1	= Represents the 10% relative precision

The following assumptions are made to exemplify the sample size calculation for the parameters.

1. An overview of the estimated sample sizes for a hypothetical population of 100,000 ICS units applying a level of 90/10 is provided below. It is likely that all the sample frames for each parameter will include fewer than 100,000 ICS in the first monitoring period, so this is a conservative approach. Hence, population size, N, is taken as 100,000 households/ICS (Assuming one ICS for one household for reference calculation).
2. It is expected at least 80% of ICS still in operation, hence the expected proportion p is taken as 0.8.

Sample size calculation:

The calculation of the required sample size for each parameter in the first monitoring period is illustrated below. In all cases a conservative approach is taken, however if for any parameter the required confidence/precision is not met then the PP will randomly select an additional sample and collect further data from this sample to ensure the pooled data meet or exceed the required thresholds.

Based on the above assumptions, the resulting sampling size is calculated as:

$$\mathrm{n} \geq \frac{1.645^2 \times 100{,}000 \times 0.8(1-0.8)}{(100{,}000-1) \times 0.1^2 \times 0.8^2 + 1.645^2 \times 0.8(1-0.8)} = 67.61$$

Therefore, in this case a sample size of 68 is to be sampled from primary sampling unit.

(ii) Mean value parameter ($B_{y=1,new,i,\ survey}$)

To estimate the sample size for mean value parameter, the following equation[49] is used:

[48] Equation (1) of Appendix 2, Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities (Version 04.0)

$$n \geq \frac{1.645^2 NV}{(N-1)\times 0.1^2 + 1.645^2 \times V}$$

Where:

$$V = \left(\frac{SD}{mean}\right)^2$$

n = Sample size

N = Population size (Total number of households/ICS)

mean = Expected mean of fuel wood consumption

SD = Expected standard deviation

1.645 = Represents the 90% confidence required

0.1 = Represents the 10% relative precision

Based on the above assumptions, the sample size calculation would be

$$n \geq \frac{1.645^2 \times 100{,}000 \times \left(\frac{0.076}{0.38}\right)^2}{(100{,}000-1)\times 0.1^2 + 1.645^2 \times \left(\frac{0.076}{0.38}\right)^2} = 10.82$$

If the resulting sample size based on the above equation is smaller than 30 ICS, then as the parameter of interest is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be used.

The sample size for mean value parameter is referred to the equation below[50]:

$$n = \left(\frac{t_{n-1} \times SD}{0.1 \times mean}\right)^2$$

Where t_{n-1} is the value of the t-distribution for 90% confidence when the sample size is n. Since the sample size is not known yet, the first step is to use the value for 95% confidence when the sample is large, i.e. 1.645 and then redefine the calculation.

The calculation now need to repeat using t-value for 90% confidence. The process should be iterated until there is no change to the value of n.

[49] Equation (4) of Appendix 2, Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities (Version 04.0)

[50] Paragraph 102, Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities (version 04.0)

Sample calculation

$$n = \left(\frac{1.645 \times 0.076}{0.1 \times 0.38}\right)^2 = 10.82$$

Thus, n is rounded up to 11.

The calculation now needs to repeat using t-value for 90% confidence and n = 11

$$n = \left(\frac{2.228 \times 0.076}{0.1 \times 0.38}\right)^2 = 19.86$$

And n is rounded to 20.

The calculation now needs to repeat using t_{n-1} value for n = 20. The process should be iterated until there is no change to the value of n.

T_{20-1}	2.093
n=	17.52
Round up	18

t_{18-1}	2.110
n=	17.81
Round up	18

The repeated calculation shows that n = 18. Thus, the sample size to be sampled from each sampling unit is 18.

Since parameters $N_{y,j,j}$, and $B_{y=1,new,i,\ survey}$ share the same sampling units, PP may choose to have one common survey for these two parameters with largest number of sample size between these two parameters being chosen. Sampling more than one parameter within the same sample (household) helps reduce travel needs for monitoring and the associated costs. At the same time this approach ensures the random selection of samples for every parameter.

It is a good practice to employ oversampling at the design stage,, not only to compensate for any attrition, outliers or non-response associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved and additional sampling efforts would be required. The sample size arrived above will be adjusted upwards to account for non-responses, Project proponent will determine the appropriate non-responses rate based on previous experience.

(b) Data to be collected:

(i) Field Measurements:

To monitor the number of stoves that continue to be in use ($N_{y,i,j}$), the data collected will be a representative number of stoves in the database that are in use for the monitoring period. The method of collecting data will be field surveys of required sample size of ICS users in the database. Data will be collected from the field surveys, entered in the database and included in the monitoring report.

For monitoring $B_{y=1,new,i,\ survey}$, the data collected will be representative number of stoves in the database that were distributed within one year of start date of project implementation. The method of collecting data will be as per $B_{y=1,new,i,\ survey}$ as given in Sec 5.2 above and frequency will be once during the entire crediting period.

The table below summarizes field measurement data requirements.

Parameter	Timing (indicative)	Frequency as required by methodology	Methods to be applied	Comments on seasonal fluctuation
$\boldsymbol{N_{y,i,j}}$	Monitoring will likely occur every 12 to 24 months	Biennially	Visits to the premises, visual inspection and interview with ICS end-user.	Seasonal fluctuation is not expected to affect this parameter.
$\boldsymbol{B_{y=1,new,i,j,survey}}$	Monitoring will likely occur within the first year after installation	First year of installation	Visits to the premises, visual inspection and interview with ICS end-user.	Seasonal fluctuation is not expected to affect this parameter. However, in case the survey results show a difference, the same shall be taken into ER estimations calculations

$B_{y=1,new,I,j,survey}$ parameter determined in the first year of the introduction of the devices through measurement campaigns at representative households., accordingly the parameter table for $B_{y=1,new,I,j,survey}$ in Sec 5.2 of VCS PD updated.

Under the project activity two TLC-CQC Rocket Stoves will be installed in each of the target households, which are termed project stove 01 and project stove 02. At the time of monitoring survey, field staff asks the end user/beneficiary HH to make a pile for the total firewood required for cooking in a day for all the stoves available in his/her house.

Further user was asked to extract and make the piles(stock 01 & stock 02) for the wood required for the project stove 1 and project stove 2 separately from that pile and weigh[51] both the piles. The field staff visit the HH on the next day[52] and weigh the balance of firewood left in the stocks and also make the note of any new wood fuel if any added to respective stock and asks the end user to show the quantity of wood added to respective stock and measure the same too for determining net firewood consumed by the project stoves of the household.

The measurement values will record in the survey forms and in the spreadsheet. Therefore, firewood consumed for each project stove(s)determined through measurement campaigns .

Prior to the measurement the end users(test samples) are instructed by the field staff

- that wood from this stock(Pile 01 & pile 02) shall not be used in baseline stove.
- end user to maintain average cooking habit.
- end user to use wood for Project stove 1 only from stock 1 and for project stove 2 they should use wood only from stock 2.

Reference	Stock 01	Stock 02
Day 0	$WPS1_{Day0}$	$WPS2_{Day1}$
Day 1	$WPS1_{Day1}$	$WPS2_{Day1}$
Fuel consumed *(Difference of the Day0 and Day 1 measured stock quantity)*	$WPS1_{Day0}$ - $WPS1_{Day1}$	$WPS2_{Day0}$ _ $WPS2_{Day1}$

WPS1- Weighment of the project stove 01_fuelwood stock ; WPS2- Weighment pf the project stove 02_fuelwood stock

(ii) Data Archiving

Hard copies of the surveys will be kept, and the registration database will have back up. Original stove purchase contracts or other means of acceptance by the users will be stored in the main office for the project proponent or coordinating entity. A back-up of the registration database will also be stored on an electronic medium by the PP. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of VCUs for the project activity, whichever is later.

(ii) Quality Assurance/Quality Control

The project proponent will apply measures to ensure the required confidence/precision for each sampled parameter is met, allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to each parameter will depend on the cost of each data collection approach and logistics required.

[51] weighing scales used for measuring the fuel wood.

[52] Next day(day 01) i.e., 24 hrs and mostly the same time as visited during the previous day (Day 0)

The project proponent will determine the most effective measure for each parameter from the following list: (illustrated using a required sample size of 20 and an effect of non-response of 2 to 4 ICS):

- Oversampling: Randomly draw a sample of minimum 24 ICS and collect data from each
- Buffer Group: Randomly draw a sample of at minimum 24 ICS and collect data from only 22 ICS. If this would not result in the required sample size data would be collected from the additional 2 ICS that were selected in the sample.
- Draw an additional sample: Randomly draw a sample of 22 ICS and collect data from these. If the required sample size is not achieved, an additional sample of 2 elements will be drawn and included in the sample.
-
- If precision required is not achieved by reliability check, use the lower confidence bound of estimates of the monitored parameter or with a conservative approach according to the parameter definitions

The PP may choose to stop monitoring a particular parameter once the required level of confidence/precision has been reached, as long as the calculated minimum number of samples has been achieved.

As an example, the following steps could logically be followed for the case of applying a 30% buffer:

a. Visit first 10% of premises required for the 30% buffer. If the number of responses is sufficient to achieve the required reliability level, then stop sampling.
b. If step 1 is not sufficient to achieve the required reliability level, then visit the next 10% of premises (increases the additional sampling to 20% of the 30% buffer). If this additional sampling is sufficient, then stop sampling.
c. If step 2 is not sufficient to achieve the required reliability level, then complete the final 10% of the additional sampling buffer (bringing the total to 30%).

The sampling plan has the following procedures in place to ensure good quality data. The project proponent will ensure that field personnel have reviewed, understand, and have agreed to follow the monitoring plan procedures, including provisions for maximizing response rates, documenting out-of-population cases, refusals and other sources of non-response. A quality control and assurance strategy will be documented. Quality control and assurance strategies include addressing non-sampling errors, such as non-response or bias from interviewer. The monitoring plan will explain how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household will be chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires will be used.

The calculation of the sample size will be carried out using estimates for parameter proportions, mean values, and standard deviations, as the actual characteristics of the population/sampling frame are unknown. In order to ensure the quality of the sampling results, the PP can draw on the provisions for reliability calculations as provided by the Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities (version 04). In the event that the sampling results do not fulfil the required level of confidence and precision, the PP can undertake additional samples. If the reliability is still not sufficient after additional samples or other measures, the sampling may be repeated with an increased sample size. Alternatively, the PP may choose to apply the lower bound or higher bound according to the more conservative approach.

(iii) Analysis

The project proponent will manage a project database that includes the following data that can be directly attributable to the project, thereby allowing unambiguous determination of the emission reductions attributable to each project:

- A list of households participating in project activity, including name, community/location, distribution/installation date and unique serial number.
- Where replacements are made, assurance that the efficiency of the new ICS is similar to the specified.

Data obtained from the samples will be used to estimate proportions and mean values for the parameters described above. The values will then be factored into the emissions reduction calculations and result in the request for issuance of VCUs. The stoves that are not in use will be excluded from emissions reductions calculations and will not be counted towards the total number of ICS in operation during the monitoring period.

<u>(c) Implementation</u>

Sampling for the purpose of emission reduction calculation and elaboration of the monitoring report will occur at the end of each monitoring period. This sampling will be conducted by trained personal either part of project implementer or pp team or an experienced third-party entity. The maximum length of one monitoring period will be two years (duration, not calendar years), The project proponent will be responsible for managing household data collection and entry into the project database. Field personnel will receive training on how to properly deal with surveying techniques and reduce errors. The project database will record the start and end dates of each monitoring period and record the emission reductions attributable to each monitoring period. Appropriate record-keeping procedures will be implemented to ensure that each monitoring period data set can be transparently attributed to its corresponding project, preventing any occurrences of double counting. An internal review of the project database will be able to determine the current status of project activity — the duration of previous monitoring periods, the households delivering monitoring data, and current verification activities.

(i) Assessment for Leakage

The applied methodology VMR0006, version 1.1 (section 8.3) provides a net to gross adjustment factor of 0.95 to account for leakages, hence the surveys are not required to determine leakage.

All ICS in the project will be newly manufactured/assembled or newly installed.

(ii) Monitoring Reporting

The project proponent will assess all monitoring data and produce a monitoring report for the VVB to verify corresponding to the preceding monitoring period. This report will present the data relating to the emission reductions generated by those batches during the monitoring period.

APPENDIX 1

A) **Different mediums which were used for inviting stakeholder's opinion on the project**

1) **CQC- Website Notice for local stakeholders' consultation**

cquestcapital.com/contact/stakeholder-feedback/

Local Stakeholders Consultation Meeting – Installation of High-Efficiency Wood-Burning Cookstoves in South Africa.

C-Quest Capital, as a Project Developer, would like to request local stakeholders in South Africa to provide feedback on their carbon finance Project, "Installation of high-efficiency wood-burning cookstoves in South Africa." The project will be developed under the Verified Carbon Standard (VCS) and The Sustainable Development Verified Impact Standard (SD VISta).

The project aims to provide clean energy access to households through the distribution of fuel-efficient cookstoves by engaging the local community. The project will contribute to the sustainable development goals through decreased demand for wood fuel, reduced deforestation, reduced drudgery in wood gathering and cooking time, increased skills and employment for the local community, and improved health through reduced household air pollution.

Carbon Finance will be used to increase the feasibility of this project as it will help in providing clean energy access to households that lack awareness, required resources, and technical know-how.

Feedback on the proposed program will be collected after the completion of the Local Stakeholder Consultation (LSC) meeting from the participants on the same day, scheduled on 17th May 2023.

Feedback can also be provided to us through the Feedback form below, from 10th May 2023 to 24th May 2023.

View Non-technical Summary
View Feedback Form

2) Newspaper advertisement for Local stakeholders' consultation meeting

Wednesday May 10, 2023 | Mpumalanga News

CLASSIFIEDS 9

Mpumalanga News

Classifieds

SERVICES

0303 Beauty, Health & Fitness
0306 Bridal
0309 Building Consultants / Designs / Plans
0312 Catering
0315 Cellphones
0318 Childcare
0321 Cleaning
0324 Computers
0327 Curtaining & Blinds
0330 Dressmaking
0331 Dancing Classes
0333 Driving Schools
0336 Entertainment Party Planning
0337 Farming
0339 Financial / Loans
0342 For Hire
0345 Furniture Restorations
0346 Functions
0348 General Repairs
0350 Hobbies / Crafts
0351 Home Industry
0354 House-sitters
0355 Insurance
0357 Locksmiths
0360 Miscellaneous
0363 Personal Services
0366 Pest Control
0369 Pet Services / Accommodation
0372 Photography

PROPERTY TO LET

0505 Accommodation / Rooms to let
0510 Farms & Plots
0511 Domestic Accommodation
0515 Flats/ Units
0518 Farm Cottages
0519 Factories / Industries / Workshops
0520 Garden Flats / Cottages
0526 House Sitters
0524 Holiday Accommodation
0525 Houses
0530 Industrial Premises
0535 Offices / Shops / Business Premises
0536 Prime Properties
0540 Retirement Accommodation / Villages
0545 Storage / Parking Facilities
0549 Timeshare to Let
0550 Town houses / Duplexes / Simplexes / Cluster Homes
0555 Wanted to Rent
0560 Miscellaneous

LEGALS

0905 Auctioneers
0910 Public & Legal Notices
0915 Sales In Execution
0920 Tenders
0925 Estates
0930 Liquidations
0935 Town Planning
0940 General

0910 PUBLIC / LEGAL NOTICES

NOTICE
INVITATION FOR PUBLIC COMMENTS IN APPLYING FOR A LIQUOR LICENCE IN TERMS OF SECTION 35(2)(a) OF THE MPUMALANGA LIQUOR LICENCING ACT, 2006
PERSONAL DETAILS
I, ROBIN L. MOIR, ID: 800413 5049 082, an adult male, hereby invite written public comments concerning my application for a liquor licence to the Mpumalanga Liquor Authority to trade under the name of MUMO RESORT (PTY) LTD. I make this application on behalf of myself.
LICENCE TYPE
a) The retail sale of liquor for consumption on the premises where the liquor is sold.
BUSINESS PREMISES
Physical address: 1 BARNARD STREET, BADPLAAS, EMANZANA, 1190, being an address in the Republic of South Africa and situated within the

PUBLIC/ LEGAL NOTICES
TOWN PLANNING
LIQUIDATIONS
SALE IN EXECUTION
ESTATES

CQuest Capital

LOCAL STAKEHOLDER CONSULTATION MEETING
"INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN SOUTH AFRICA"

C-Quest Capital is developing a project titled "Installation of high efficiency wood burning cookstoves in South Africa" under the Verified Carbon Standard (VCS) and Sustainable Development Verified Impact Standard (SDVISta).

The project aims to provide clean energy access to households by distribution of fuel-efficient improved cookstoves (ICS) in rural and semi-urban communities of South Africa.

In this regard, C-Quest Capital extends an invitation to all stakeholders as well as interested parties for participating in the Local Stakeholder Consultation scheduled to take place on 17 May 2023, starting at 9:30 AM at Hazyview Sun (R40 Road 38km after White River, Hazyview, 1242) to discuss the project design and its impact on sustainable development, as well as to solicit your valuable feedback.

For registration, please contact us on +27 (0) 66 383 0325 or write to us at: cqc_csat@cquestcapital.com. You may also contact us on the above for any queries or comments.

CQuest Capital

NHLANGANO WA VAAKA TIKO

"KU AKA SWITOFU SWO TIRHISA TIHUNYI HI VUSWIKOTI EAFRIKA DZONGA"

C-Quest Capital (CQC) yi kulisa/kuhumelerisa ntirho wo "aka switofu swo tirhisa tihunyi eAfrika Dzonga" ku hunguta musi no cinca matirhiselo ya tihunyi. Ntirho wo tisa kuhunguta kuchakisa ka mimoya hiku tisa switofu swo antswisa e matikoxikaya eAfrika Dzonga.

Kuya hi C-Quest Capital yi rhamba hinkwavo vaaka tiko xikan'we nava n'watipolitiki kuva va nghenelela eka nhlangano lowu vekiweke hi ndlela leyi landzelaka **17 May 2023 (9:30 AM) endhawini ya Hazyview Sun (R40 Road 38km after, white River, Hazyview, 1242)** kuvulavula hi maendlelo ya ntirho na nkoka wa kucinca, na kutiva switandzhaku swa kona.

Ku tsarisa, tihlanganisi na hina ka **+27 (0) 66 383 0325** kumbe tsalela eka hina ka **cqc_csat@cquestcapital.com**
Ungathlela utihlanganisa na hina hi swivutiso kumbe mavonelo/kutatisa.

LOCAL STAKEHOLDER CONSULTATION MEETING

"INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN SOUTH AFRICA"

C-Quest Capital is developing a project titled "Installation of high efficiency wood burning cookstoves in South Africa" under the Verified Carbon Standard (VCS) and Sustainable Development Verified Impact Standard (SDVISta). The project aims to provide clean energy access to households by distribution of fuel-efficient improved cookstoves (ICS) in rural and semi-urban communities of South Africa.

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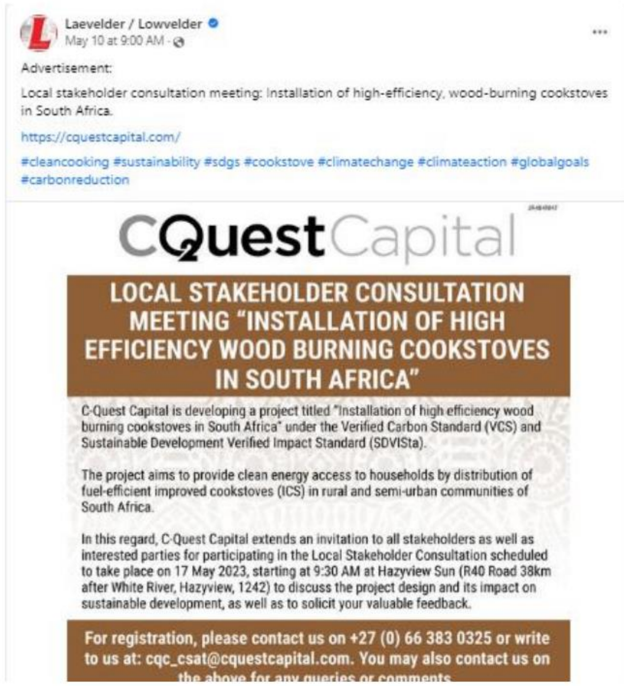
You may also contact us on the above for any queries or comments

FACEBOOK STATS



19 076
Reach

217
Link Clicks

3) Email invitation and invitee list

Delete | Archive | Report | Reply | Reply all | Forward | Read / Unread | Categorize | Flag / Unflag | Assign polic

Consultation meeting with local stakeholders for the project "Installation of high efficiency wood burning cookstoves in South Africa"

C

CQC_CSAT

To: Vijay Kumar Machcha

Tue 5/2/2023 9:12 PM

Cc: Koganti Narendra Nath; CQC_CSAT

Bcc: makhuvelest@agric.limpopo.gov.za; mabundabp@agric.limpopo.gov.za; sibanyonipc@mpg.gov.za; nmbedu@mpg.gov.za; shabanguzj@mpg.gov.za; nyathikazibw@mpg.gov.za; nyathikazibw@mpg.gov.za; goxaba@mpg.gov.za; omagagula@mpg.gov.za; goxaba@mpg.gov.za; nsylvia706@gmail.com; mushwanakj@gmail.com; nyameka.bongco@drdar.gov.za; philiswa.pukwana@drdar.gov.za; zoleka.ntondini@drdar.gov.za; nonkonzo.mnyengeza@drdar.go.za; zimasa.nduna@drdar.gov.za; sebenzileubisi@yahoo.com; gaymoreme@gmail.com; delfortmbetse@gmail.com; masingawonderboy06@gmail.com; kulanivc@gmail.com; victorncube664@gmail.com; masingawonderboy06@gmail.com; hectormoroane@yahoo.com; doriesnyathi@gmail.com; mdlamini@wrhi.ac.za; preventionz@sancalowveld.com; olga.monareng@m2m.org; maphelohbc@gmail.com; tintsaalo@grip.org.za; mothaileng@gmail.com; mpumi.mdaka@gmail.com; fortunendlovu79@gmail.com; siphwemdluli01@gmail.com; maki.lethabo@gmail.com; gladmaziba@gmail.com; kgafanekedibone@gmail.com; impangana@edc.org; timm.hoffman@uct.ac.za; lindsey.gillson@uct.ac.za; info.acdi@uct.ac.za; sharon.adams@uct.ac.za; claire.khai@uct.ac.za; lilly.sithole@ump.ac.za; nelisiwe.majola@wits.ac.za; erik@sun.ac.za; catherine@sun.ac.za; garena@conservation.org; team@africanclimatealliance.org; team@africanclimatealliance.org; diole@hpp-sa.org; Mathews@hpp-sa.org; jillian@hpp-sa.org; zimondik@hpp-sa.org; msandex@hpp-sa.org; mahungela111@gmail.com; tiyapo@hpp-sa.org; info@greencape.co.za; andre@ener-g-68africa.com; catherine@ener-g-africa.com; louise@ener-g-africa.com; schalk@ener-g-africa.com; amina@ener-g-africa.com; info@c4es.co.za; Vikash Kumar Singh <vikash@carboncheck.co.in>; Harish Sharma <harish@carboncheck.co.in>; Shalini Yadav <shalini@carboncheck.co.in>; lloyd.archer@united-purpose.org; syadav@globaloffsetresearch.com; david@cedesol.org; Sutrishna Khan; ruanparrott@earthscienceafrica.com

Non technical summary_Coo... 520 KB

Feedback Form_Cookstoves_... 35 KB

2 attachments (556 KB) Save all to OneDrive - C-Quest Capital Download all

Dear Sir/Madam,

C-Quest Capital, as part of the requirement under the process, is pleased to invite you to participate in the consultation meeting of Local Stakeholders for the project entitled 'Installation of high efficiency wood burning cookstoves in South Africa' scheduled on 17 May 2023, 9:30AM at Hazyview Sun, R40 Road 38km After, White River, Hazyview, 1242, South Africa.

The meeting agenda and details are provided below for your reference (as per South Africa time, GMT+2:00).

09:30 – 10:00 AM – Welcoming and speeches by guests
10:00 – 11:00 AM – Introduction to the Verified Carbon Standard and SDVISta
11:00 – 11:30 AM – Presentation on the project design and plan
11:30 AM onwards – Q&A and feedback session

The project aims to provide clean energy access to households through the distribution of fuel-efficient cookstoves in South Africa. The project will contribute to the sustainable development goals through decreased demand for wood fuel, reduced deforestation, reduced drudgery in wood gathering and cooking time, increased skills and employment for the local community, and improved health through reduced household air pollution.

Considering the financial implications associated with the project, carbon finance will be used to increase the feasibility and scale-up of this project. Carbon Finance will be used to increase the feasibility of this project as it will help in providing clean energy access to households that lack awareness, required resources, and technical know-how.

The proposed project will be developed under the Verified Carbon Standard (VCS) and the Sustainable Development Verified Impact Standard (SD VISta) to obtain SD VISta labelled Verified Carbon Units (VCUs). Carbon credits will be used towards providing access to clean cooking technology which is currently deficient in the target communities owing to lack of awareness, required resources, and technical know-how.

Those who would not be able to attend the in-person meeting may please join Google Meet with the links given below.

Google Meet link: https://meet.google.com/vfx-fiqc-wgz
Phone link for the meet: https://tel.meet/vfx-fiqc-wgz?pin=7403422770821

Non-technical summary and feedback form are attached with this e-mail for your consideration.

In case of any clarification or comment/suggestion regarding said projects, you can contact us at cqc_csat@cquestcapital.com.

We look forward to your participation.

Thanks, and regards.
C-Quest Capital LLC
1015 18th Street, NW, Ste 730
Washington DC, USA 20036
http://www.cquestcapital.com/

Reply | Reply all | Forward

B) Stakeholder Consultation meeting Q&A session

Name	Comment /Question	PP response
Morris Hlongwane (Humana People To People)	Who will repair the stove if it breaks?"	People from the communities may contact the CQC /their partner organizations (responsible for building the cookstoves to request repairs. In addition, a constant communication and grievance redressal mechanism has been designed and shared in the meeting for seamless access
Dimakatso Nonyane (Kruger to Canyon)	if the project was aware of a similar initiative inprogress.	Yes, the project is in the process of identifying other projects in the same area and understanding the villages and provinces involved. PP will ensure that only households with the baseline situation of three stone fired, or tripod will be identified and provided with project stoves.
Elizabeth (TASC)	if the project was aware of another project undertaking the same initiative in Limpopo and Mpumalanga.	If households are already using improved cookstoves from another organization, they will not. Be eligible to receive the project stoves as per the screening criteria determined by PP as a part of the project implementation procedure. The PP will put the best efforts to investigate the implementation areas to avoid duplication.

C) <u>Feedback from stakeholders:</u>

Feedback 01: Victor Ncube (Headman of the Cork Village) expressed gratitude to CQC for selecting theircommunity as the first village to receive improved cookstoves. He appreciated the opportunity and mentioned discussing the project with other headmen in attendance.

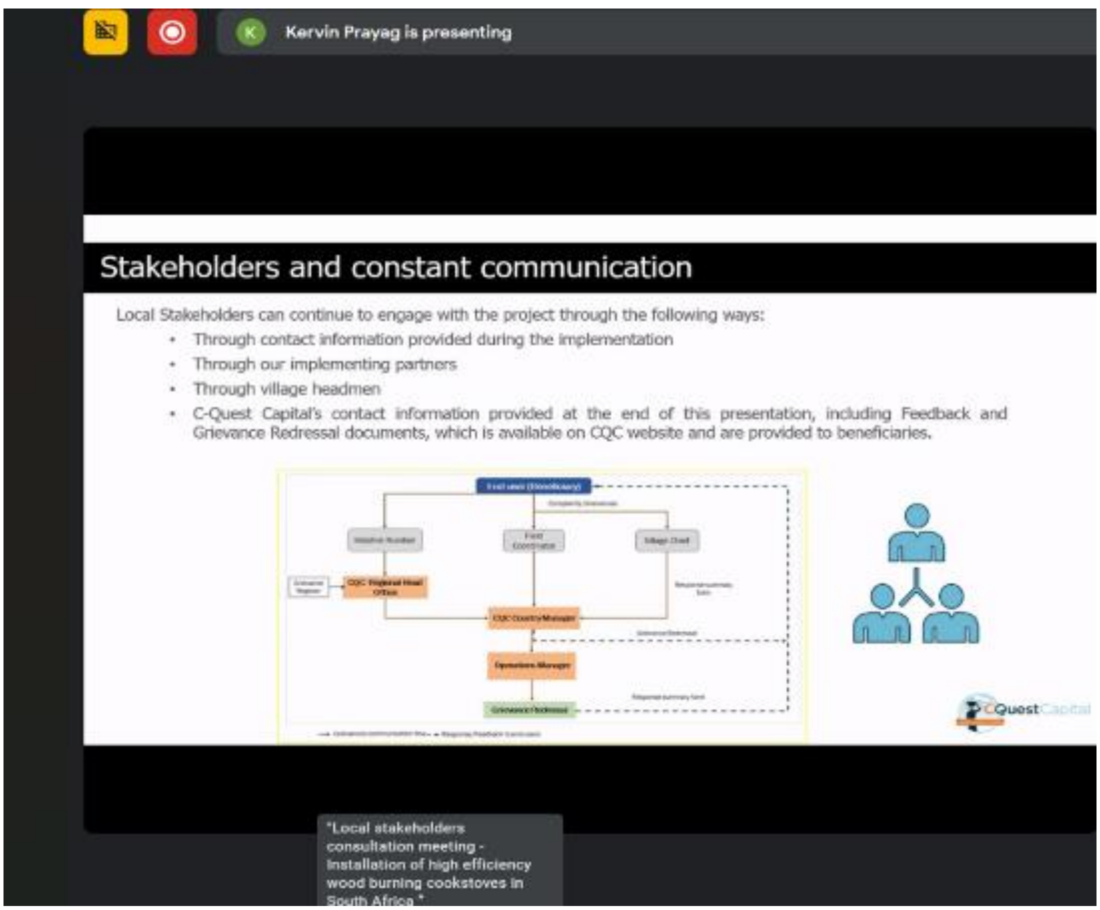
Feedback 02: Molatedi (Humana People To People) emphasized the importance and global impact of cookstoves, mentioning the benefits such as reduced air pollution and deforestation, leading to potential advantages for the tourism industry.

D) LSC meeting Photographs


VCS
Verified Carbon Standard (VCS)


EXIT


How it works?
Replacing Three stone fire cookstove with Improved Cookstove has health benefits and also results in the reduction of greenhouse gas emissions (mostly Carbon Dioxide). SD VISta labelled Verified Carbon Units will be issued by VERRA for the project.
The finance obtained from selling carbon credits is used for the sources of ICSs for the households.
Carbon finance is used for funding the Cookstove program
VCUs
Three stone fire Cookstove
Improved Cookstove
Carbon Credits
CQuestCapital
Kervin Prayag
"Local stakeholders consultation meeting - Installation of high efficiency wood burning cookstoves in South Africa "


Kervin Prayag is presenting
Stakeholders and constant communication
Local Stakeholders can continue to engage with the project through the following ways:
Through contact information provided during the implementation
Through our implementing partners
Through village headmen
C-Quest Capital's contact information provided at the end of this presentation, including Feedback and Grievance Redressal documents, which is available on CQC website and are provided to beneficiaries.
CQuestCapital
"Local stakeholders consultation meeting - Installation of high efficiency wood burning cookstoves in South Africa "

E) Sample feedback form

The following is the sample of evaluation form that the participants were requested to fill. The evaluation forms were printed in two different languages, viz Tsonga and English.

FEEDBACK FORM/MBUYELO WA XIVUMBEKO
INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN SOUTH AFRICA/KU VEKIWA KA SWILO SWALE HENHLA SWO TIRHEKA KU TIRHISA RIHUNYI EKA SWITOFU SWO SWEKA EAFRIKA DZONGA

Respondent Information/Muvutisiwa marungula:

Name/Vito:	
Organisation Name/Vito ra Nhlangano:	
Occupation/Ntirho:	
Designation/Ntirho:	
Contact number and Email/ Nomboro ya vutihlanganisi:	

1. What are the benefits you expect as a result of the Project Activity (PA) implementation?
   Yini mbuyelo lowu ngawu langutela hi swita ndzaku swa mintirho ya phurojeke leswinga tisiwa?

   ☐ It will reduce the amount of wood required for cooking.
   Swita hunguta matirhiselo ya tihunyi ta ku sweka
   ☐ It will reduce smoke and indoor air pollution, and results in health benefits.
   Swita hunguta musi na machakiselo ya mimoya makaya, na vuyerisa ku kumiwa aihanyu.
   ☐ It will reduce the drudgery in the wood collection and cooking activities.
   Swita hunguta ku tsemiwa ka tihunyi hixitalo na mintirho yo sweka.
   ☐ It will reduce costs and improve overall productivity and quality of life
   swita hunguta hakelo no andisa mbuyelo na rihanyu ra kahle.
   ☐ No benefits are envisaged.
   Akuna swita ndzaku swinga languteriwa.

2. Do you think the project will be beneficial for the end users?
   U hleketa leswaku phurojeke yita leswaku phurojeke yita vuyeriwa hi swita ndzaku?

   ☐ Yes/Ina ☐ No/E-e

3. Are the project design and implementation plan clearly explained?
   Xana vukhavisi na kunghenisiwa ka swirho swa kungu swi hlamuseriwe hi ndlela ya kahle?

   ☐ Yes/Ina ☐ No/E-e

4. Are the project feedback and grievance redressal procedures clearly explained?
   Mbuyelo na xivilelo xa phurojeke swi ngava swi hlamuseriwe swi twakala?

   ☐ Yes/Ina ☐ No/Ina

5. Do you think the project stoves will have a positive impact on the health of women and children?
   U hleketa kuri phurojeke ya switofu?

☐ Yes/Ina ☐ No/E-e

6. Do you think the project will have a positive impact on the forests of the South Africa?
   U hleketa kuri phurojeke yitava na swita ndzhaku swa kahle eka makwati eAfrika Dzonga?

   ☐ Yes/Ina ☐ No/E-e
7. Would you like to participate in this Programme?
   wa tsakela ku nghenelela ka nongonoko?

   ☐ Yes/Ina ☐ NoE-e
8. Are you aware about any other similar ongoing projects in your area?
   Awulemuka swin"wana mayelana na leswi fambisaka hi phurojeke?

   ☐ Yes ☐ No
9. Do you have any inputs, concerns, or suggestions on the project design or implementation plan?
   Kungava kuri na leswi u lavaka kuswi nghenisaka, swivilelo kumbe ndzinganyeto ka leswi vekiweke hi phurojeke kumbe ku tirhisiwa kungu?

   ____

   ____

10. What is your impression of the project results, outcomes, and impacts (both positive/negative)?
    I yini swi nga ku tsakisa hi swita ndzaku swa phurojeke, mbuyelo na ntshikelelo (ka swa kahle/ na leswo biha)?

    ____

    ____

** You may use the back of this paper or request for additional sheets if the space is insufficient.*

U nga tirhisa endzhaku ka phepha kumbe ku kombela ku tatiseriwa maphepha ya swiphepherhele loko ku hava ndhawu yo tsalela

APPENDIX 2

The screenshots of the public notice[53] to avoid double claiming of Scope 3 Emissions under this grouped project are provided below:

cquestcapital.com/latest/public-notices/

Public Notice for VCS Project 3941

"Installation of High Efficiency Wood Burning Cookstoves in South Africa"

This is to bring to your kind notice that C-Quest Capital CR Stoves Private Limited is implementing a grouped project titled "Installation of High Efficiency Wood Burning Cookstoves in South Africa" (VCS ID 3941). It involves distribution of Improved Cook Stoves (ICS) to Households in Republic of South Africa. For this purpose, Single pot TLC CQC Rocket stove will be distributed to households. Stove Metal parts will be procured from Ener-G-Africa (EGA).

Verified Carbon Units (VCUs) may be issued for the Greenhouse Gas emission reduction and removals for which C-Quest Capital CR Stoves Private Limited will be claiming credits under VERRA. The ownership of these credits lies exclusively with C-Quest Capital CR Stoves Private Limited

This notice intends to apprise you about the project, to avoid any potential risk of double claiming of Scope 3 emissions under this grouped project.

[53] https://cquestcapital.com/latest/public-notices/